The Maths of Predators and Preys

Don’t get Eaten by the Wolf

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TU Dublin (Lecturer in Mathematics and Statistics)

frontiers FOR YOUNG MINDS

MATHEMATICS
Published: 24 December 2021
doi: 10.3389/frym.2021.651131

THE CIRCLE OF LIFE: THE MATHEMATICS OF PREDATOR-PREY RELATIONSHIPS

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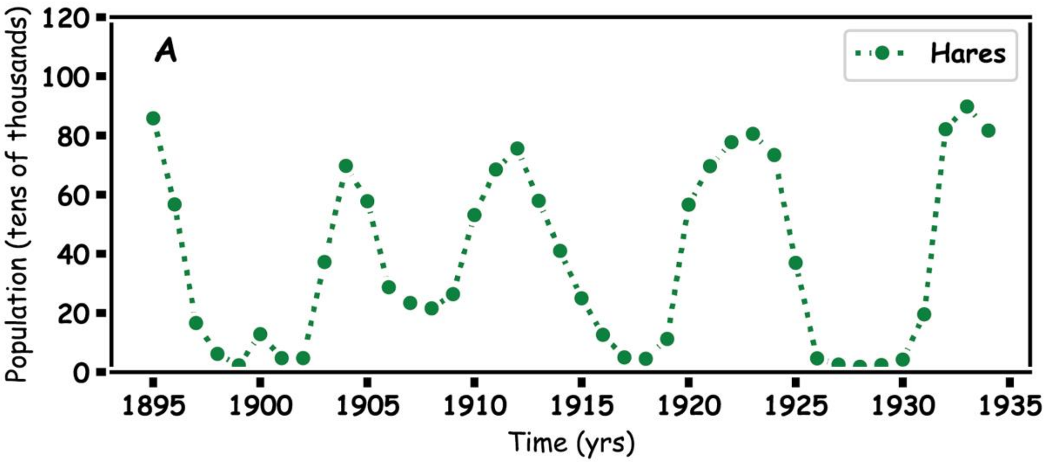
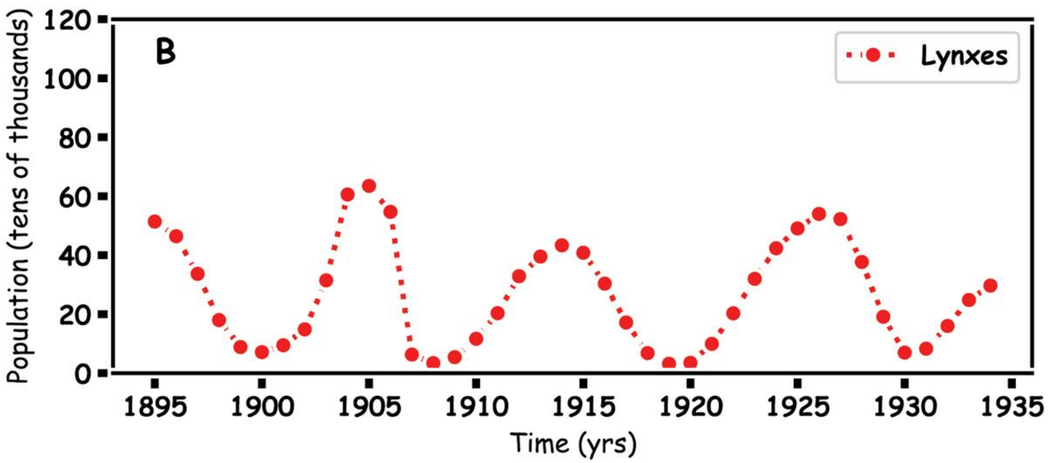
Predator Prey

Hares and Lynx

Prey Hares

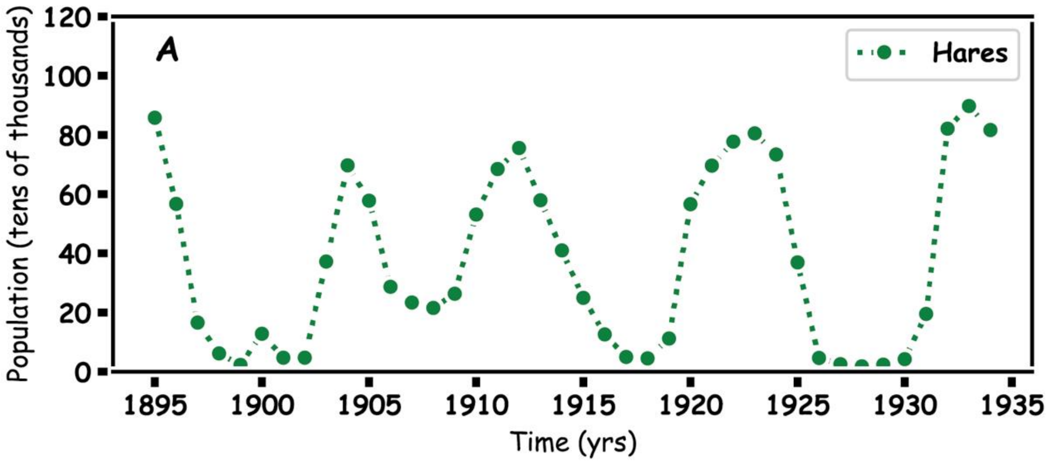

A
Hares
Population (tens of thousands)
120
100
80
60
40
20
0
1895
1900
1905
1910
1915
1920
1925
1930
1935
Time (yrs)

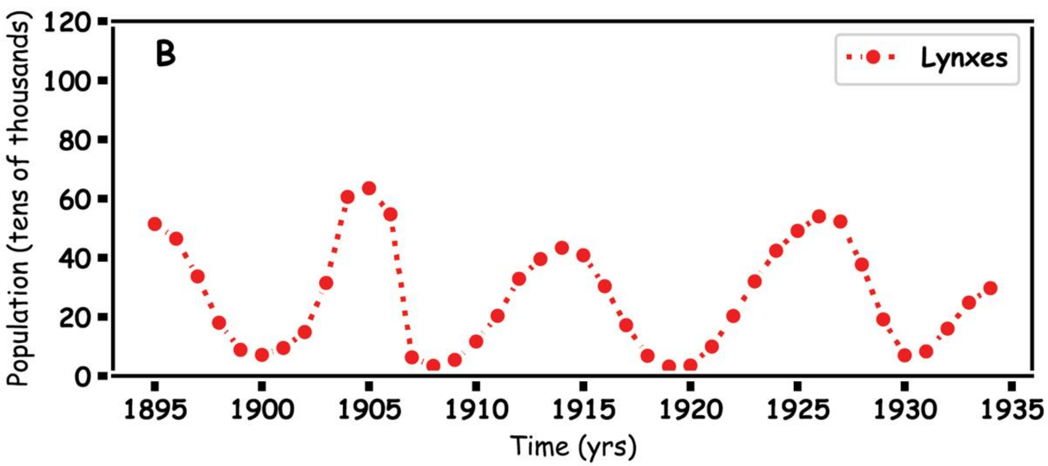
Predator



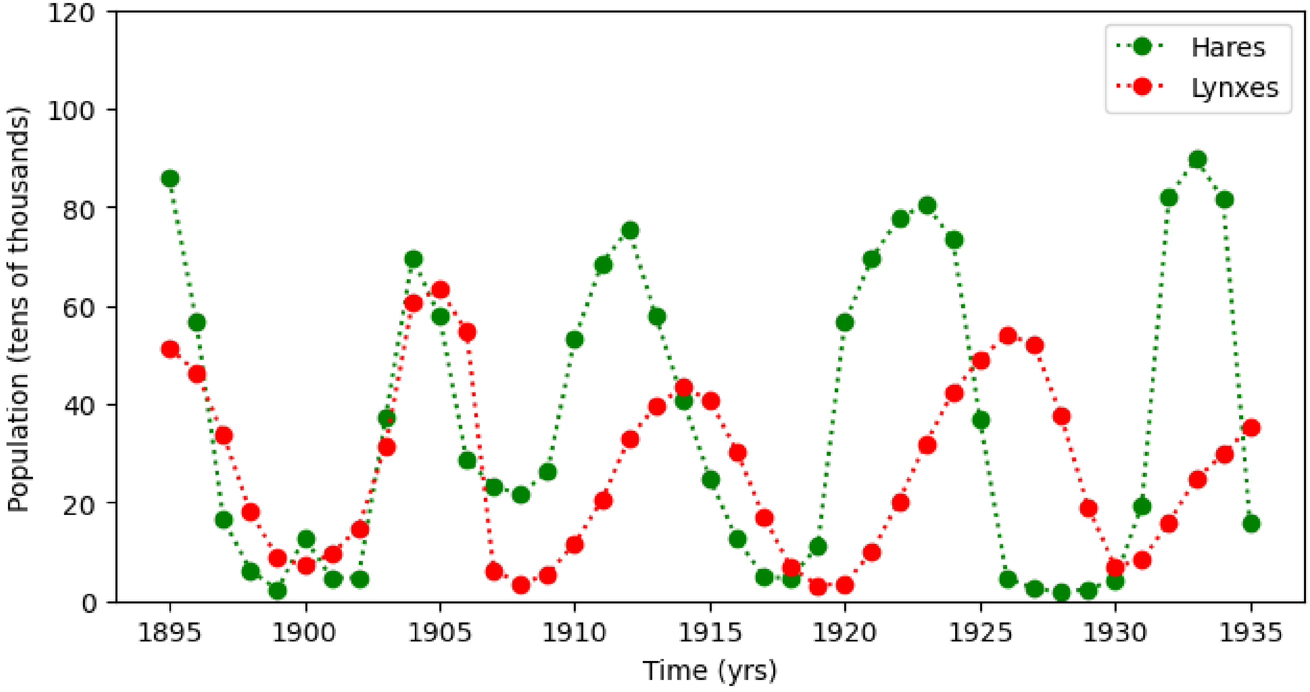
B
Lynxes
Population (tens of thousands)
0
20
40
60
80
100
120
1895
1900
1905
1910
1915
1920
1925
1930
1935
Time (yrs)

Predator and Prey

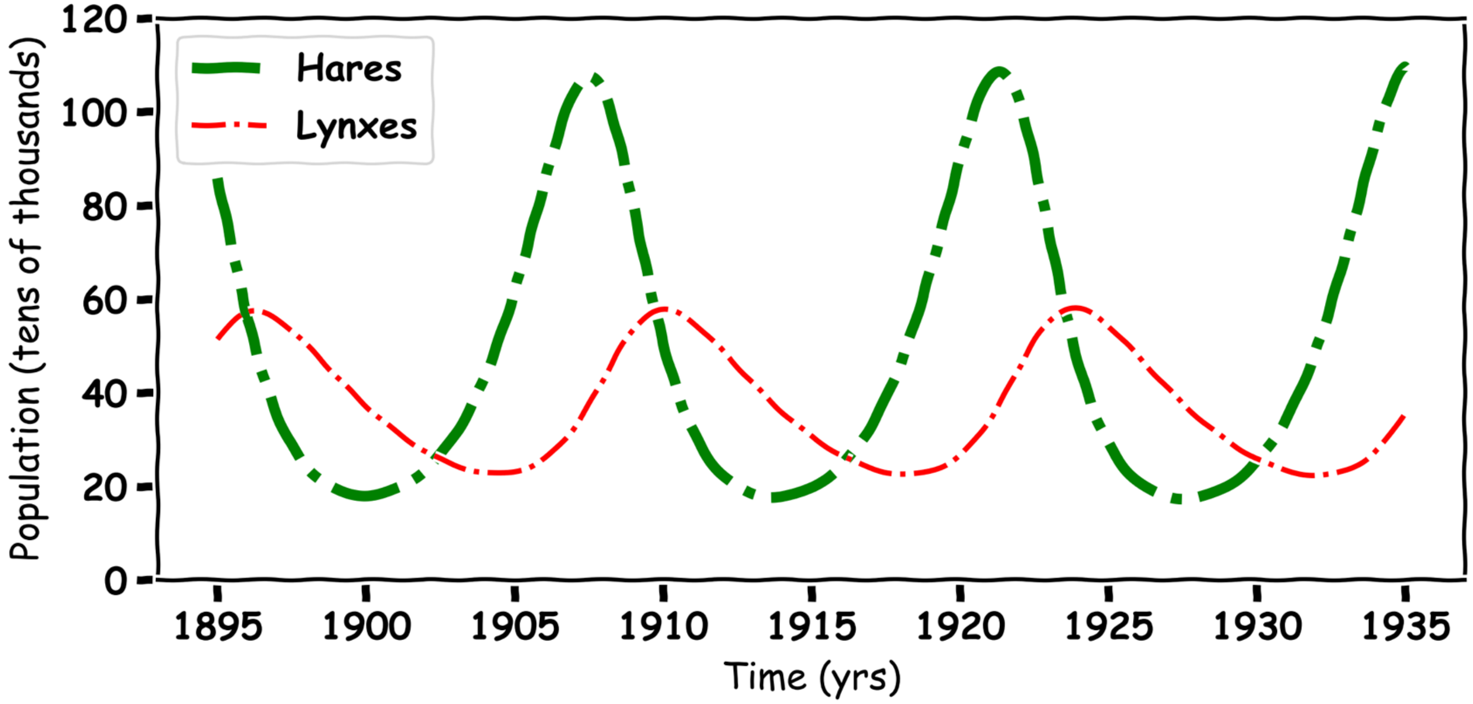




Predator and Prey



Predator and Prey



Mathematics

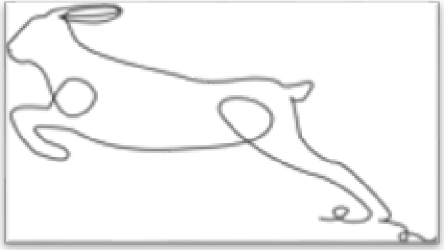
Hare Population Simulation

To make a computational model for a Hare population.
Some assumptions and simplifications must be made.

Here we make two assumptions:

1. Hares have no predators
2. Hares have unlimited food

We can now move from the real world to the simulation



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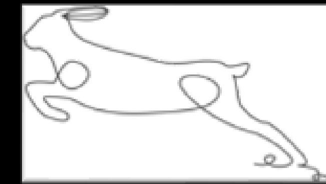
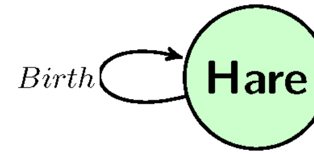
Predicting One Year in the Future

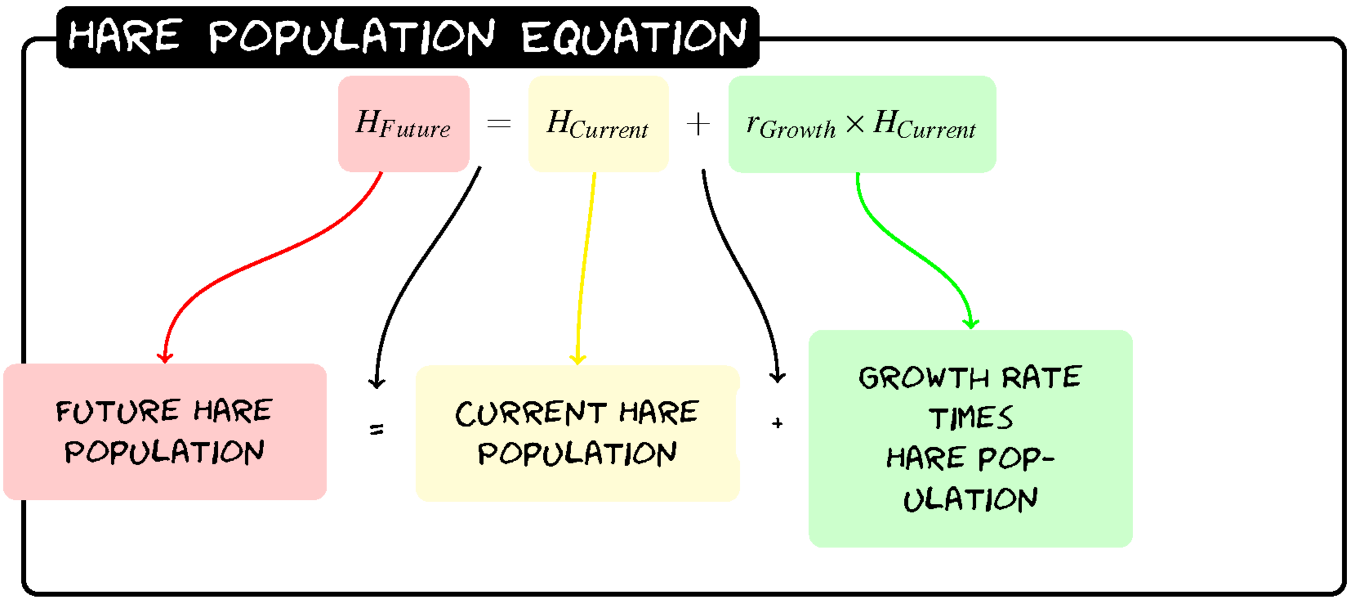
- To predict the future hare population from the current hare population.
- To do this we need a birth rate, which is the amount the Hare population will increase per year
- Now we need a simple formula using multiplication and addition to predict the future population:

current population plus the birth rate times current population

Hares







$$H_{Future} = H + 0.2(H)$$

Exercise 1: Predicting Three Years

$$\text{Future Hare Population} = \text{Current Hare Population} + \text{Birth Rate} * \text{Current Hare Population}$$

$$H = H + r_{Birth} * H$$

- Given the Hare population in Year 0 is 60 implement the formula three times to get the population in Year 3.

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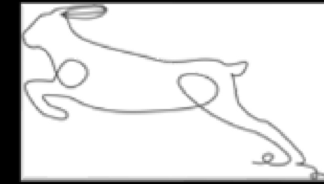
Exercise 1: Predicting Three Years

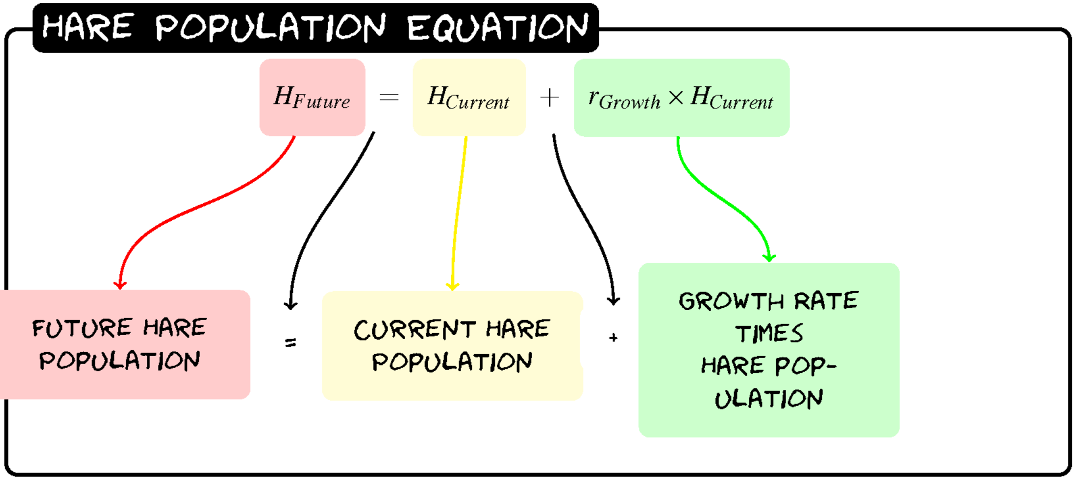
$$\text{Future Hare Population} = \text{Current Hare Population} + \text{Birth Rate} * \text{Current Hare Population}$$

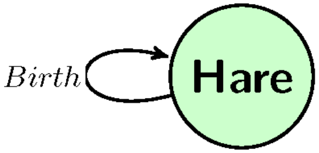
$$H = H + r_{Birth} * H$$

Year	Hare Population
0	60
1	
2	
3	

Exercise 1.1: Predicting One Year







$$H_{Future} = H + 0.2(H)$$

Year	Hare Population
0	60
1	
2	
3	

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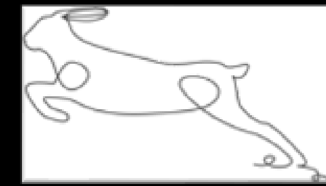
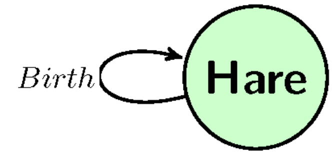
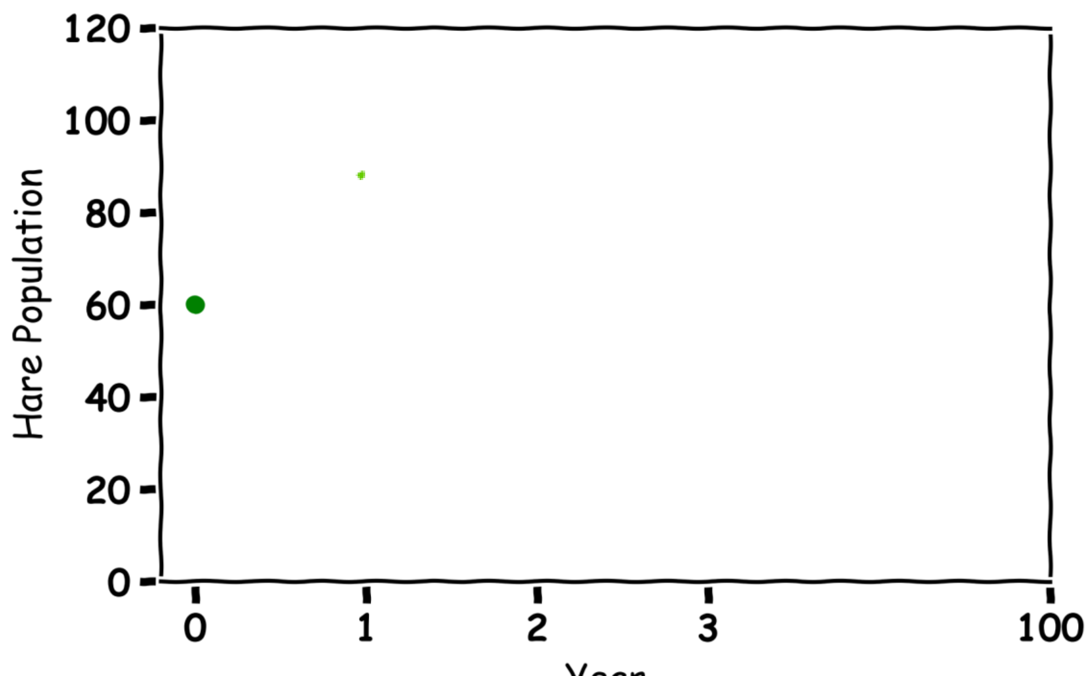
Exercise 1.1: Predicting One Year



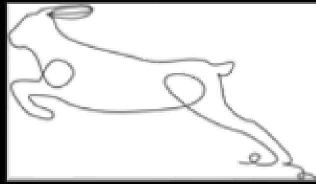
$$H_{Future}=H+0.2(H)$$



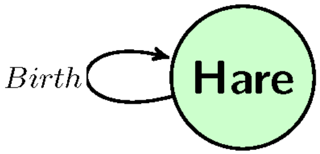
Year	Hare Population
0	60
1	60+0.2(60)=72
2	
3	



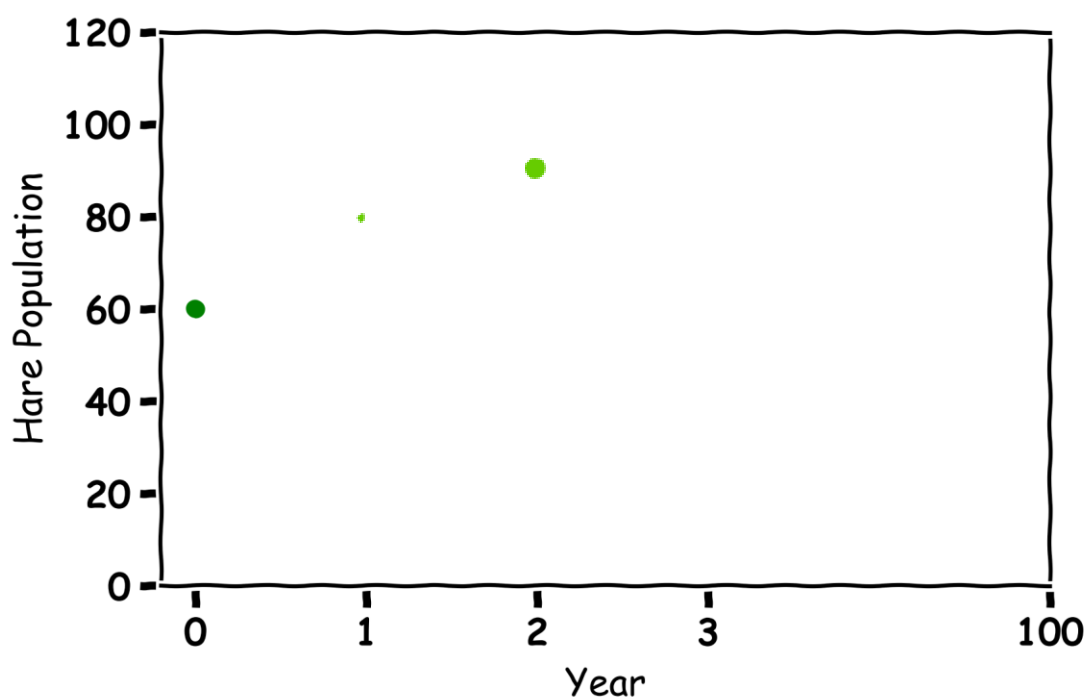
Exercise 1.2: Predicting Three Years



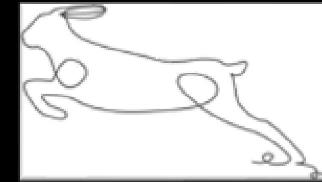
$$H_{Future} = H + 0.2(H)$$



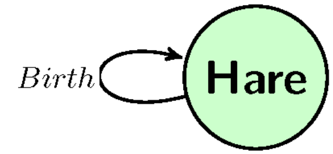
Year	Hare Population
0	60
1	60+0.2(60)=72
2	72+0.2(72)=86
3	



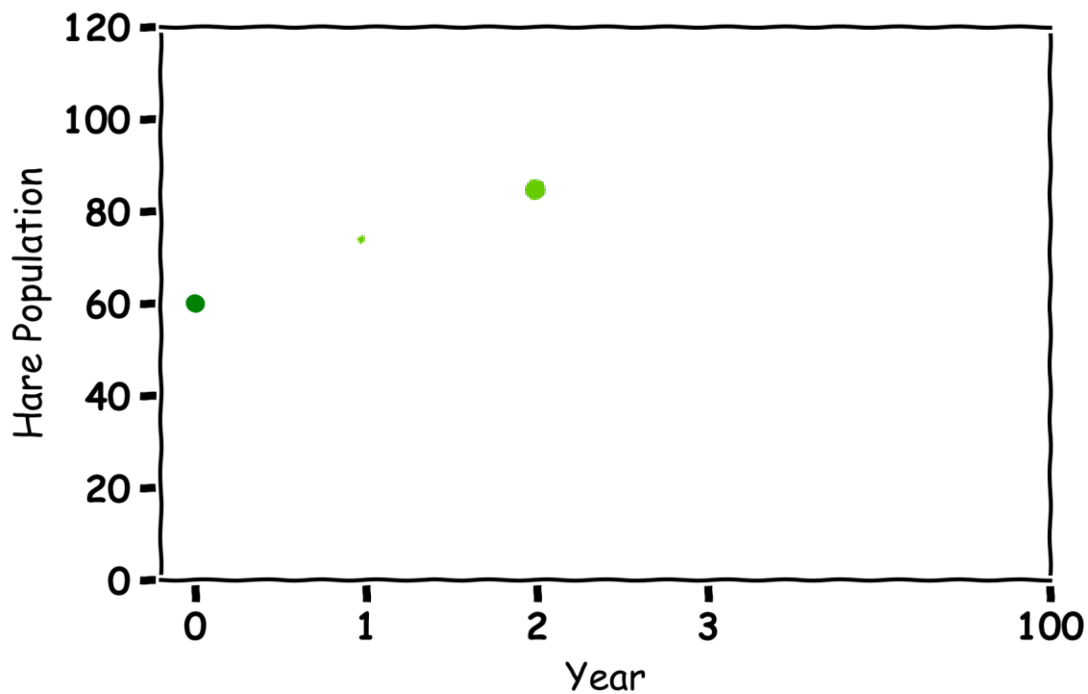
Exercise 1.2: Predicting Three Years



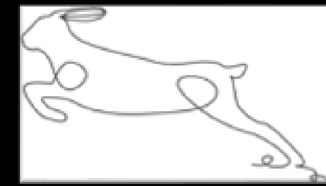
$$H_{Future}=H+0.2(H)$$



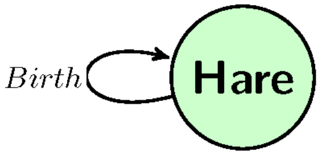
Year	Hare Population
0	60
1	60+0.2(60)=72
2	72+0.2(72)=86
3	86+0.2(86)=103

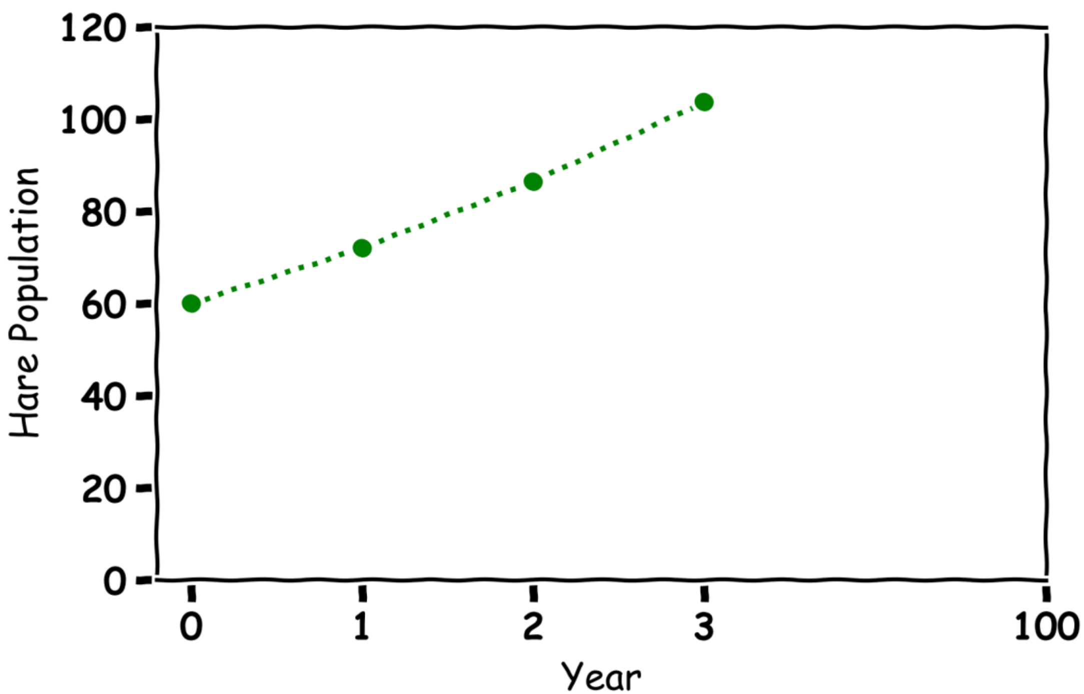


Exercise 2: Plotting Three Years

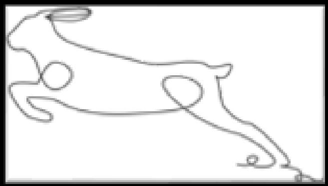


Year	Hare Population
0	60
1	72
2	86
3	103

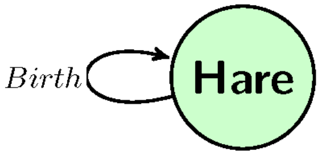


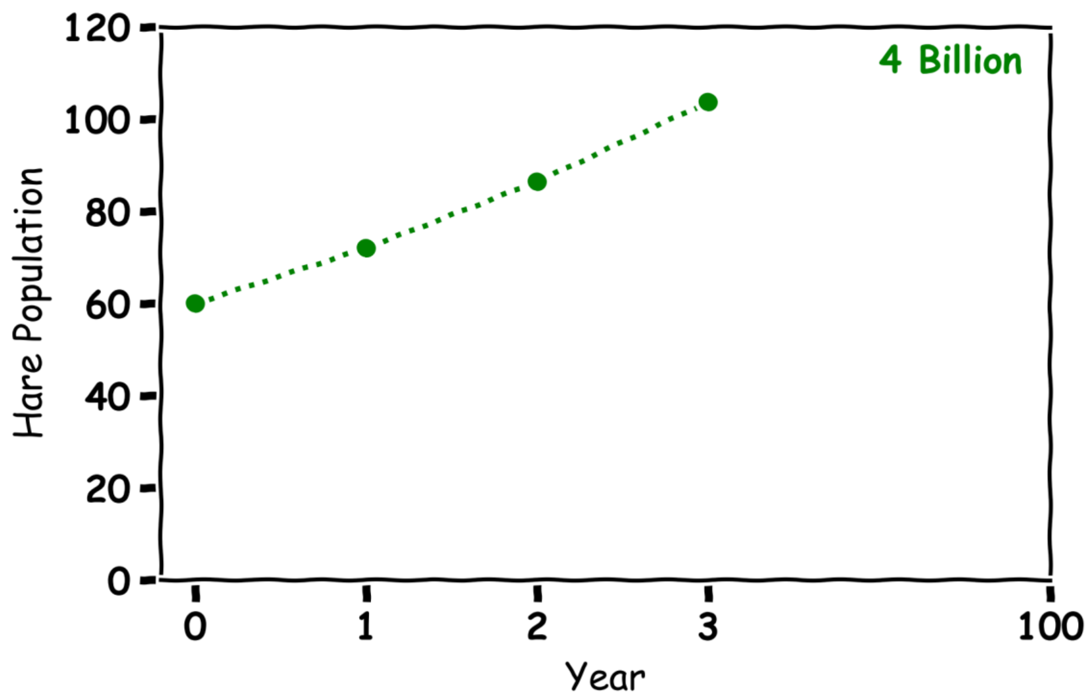


Exercise 3: Predicting 100 Years

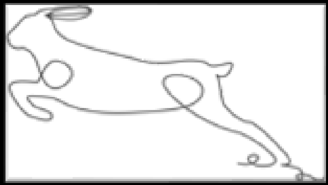


Year	Hare Population
0	60
1	72
2	86
3	103



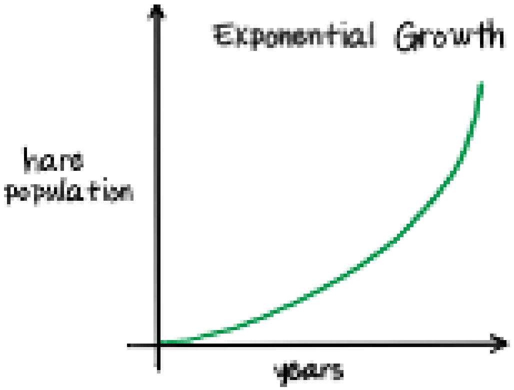


Hare Population



Ⓐ Hare Population

H_c : current hare population H_F : Future hare population



$$r_{growth} = \text{Births} - \text{Deaths}$$

$$\text{Change in hare population} = r_{growth} \times H_c$$

$$H_F = H_c + \text{Change in hare population}$$

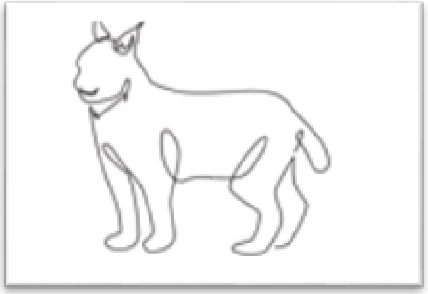
Lynx Population Simulation

To make a mathematical model for a Lynx population.
Some assumptions and simplifications must be made.

Here we make one assumptions:

1. Lynxes have no food

We can now move from the real world to the simulation

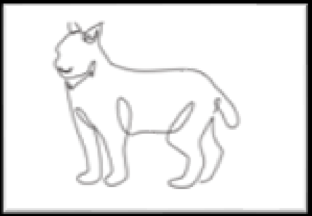


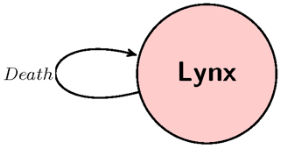
Predicting One Year in the Future

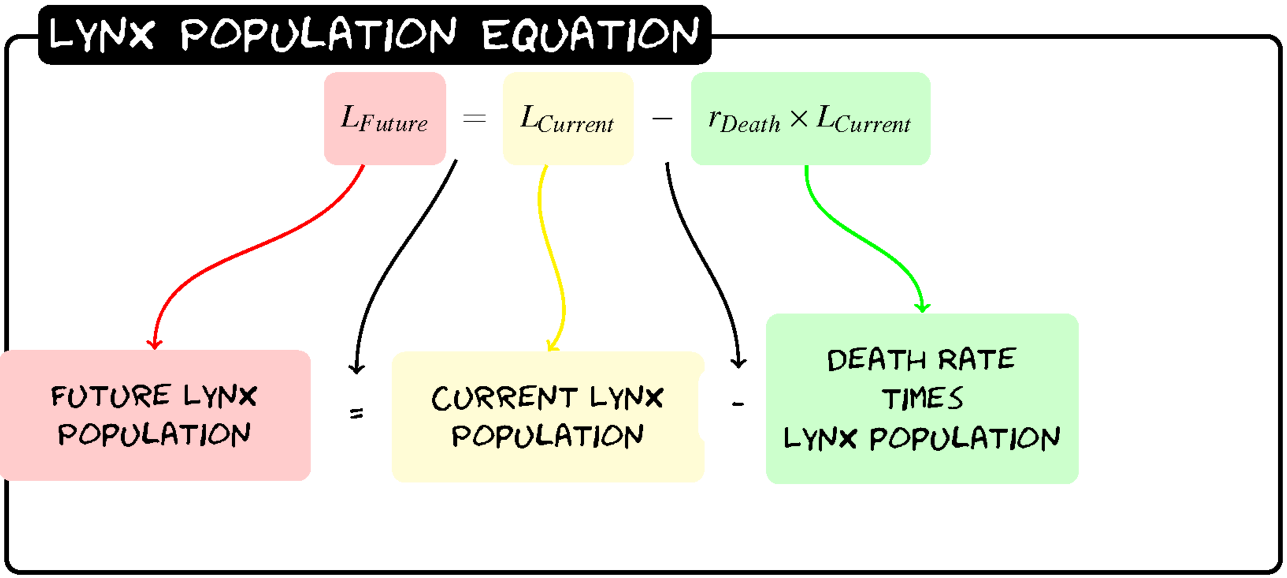
- To predict the future lynx population from the current lynx population.
- To do this we need a death rate, which is the amount the lynx population will decrease per year
- Now we need a simple formula:

current population minus the death rate times current population

Lynx Population

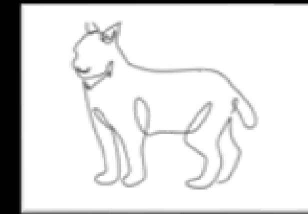


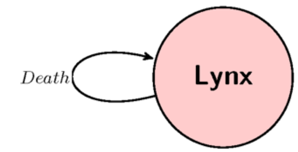




$$L_{Future} = L - 0.1(L)$$

Exercise 4.1: Predicting One Year





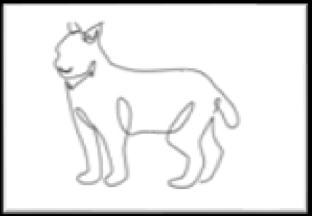
LYNX POPULATION EQUATION

$$L_{Future} = L_{Current} - r_{Death} \times L_{Current}$$

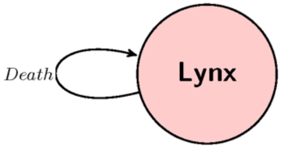
FUTURE LYNX POPULATION = CURRENT LYNX POPULATION - DEATH RATE TIMES LYNX POPULATION

Year	Lynx Population
0	30
1	
2	
3	

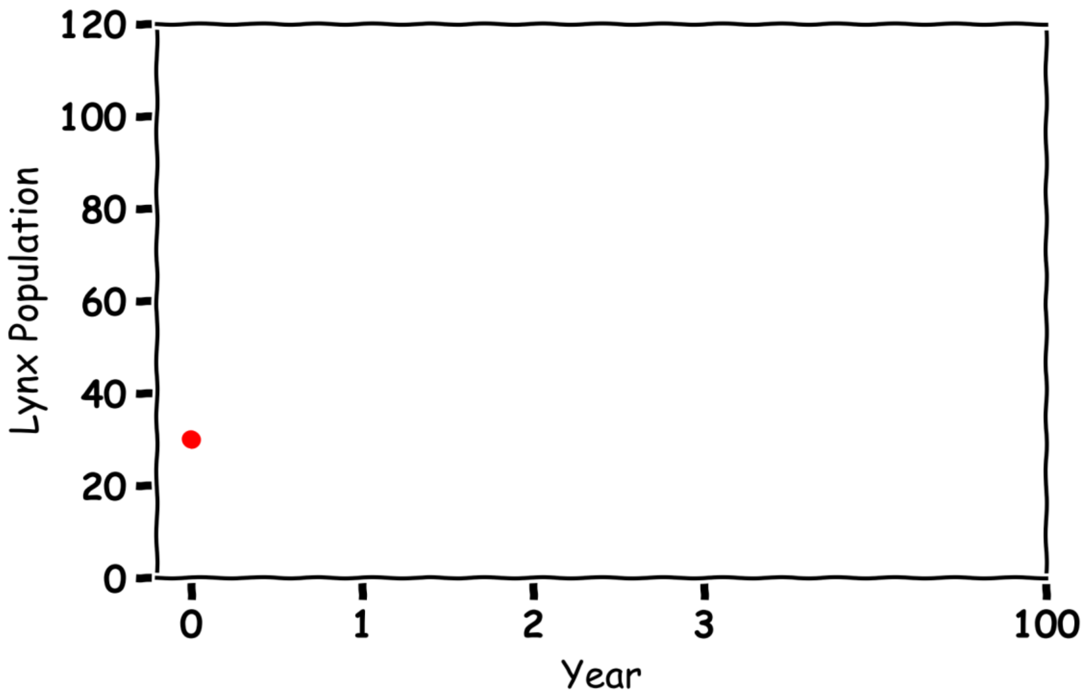
Lynx



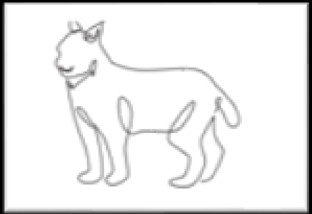
$$L_{Future} = L - 0.1(L)$$



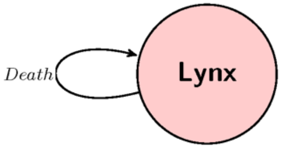
Year	Lynx Population
0	30
1	
2	
3	



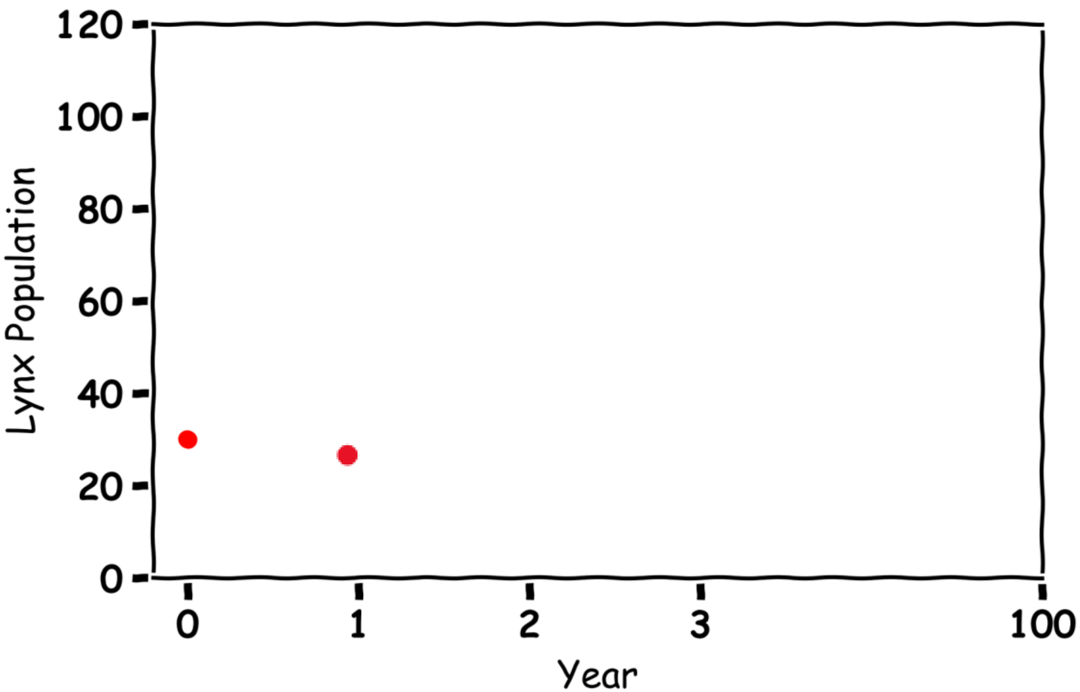
Exercise 4.1: Predicting One Year



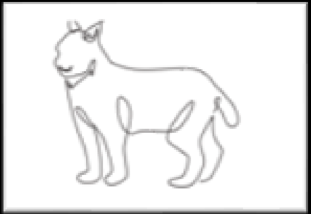
$$L_{Future} = L - 0.1(L)$$



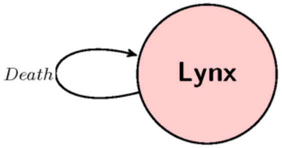
Year	Lynx Population
0	30
1	30-0.1(30)=27
2	
3	



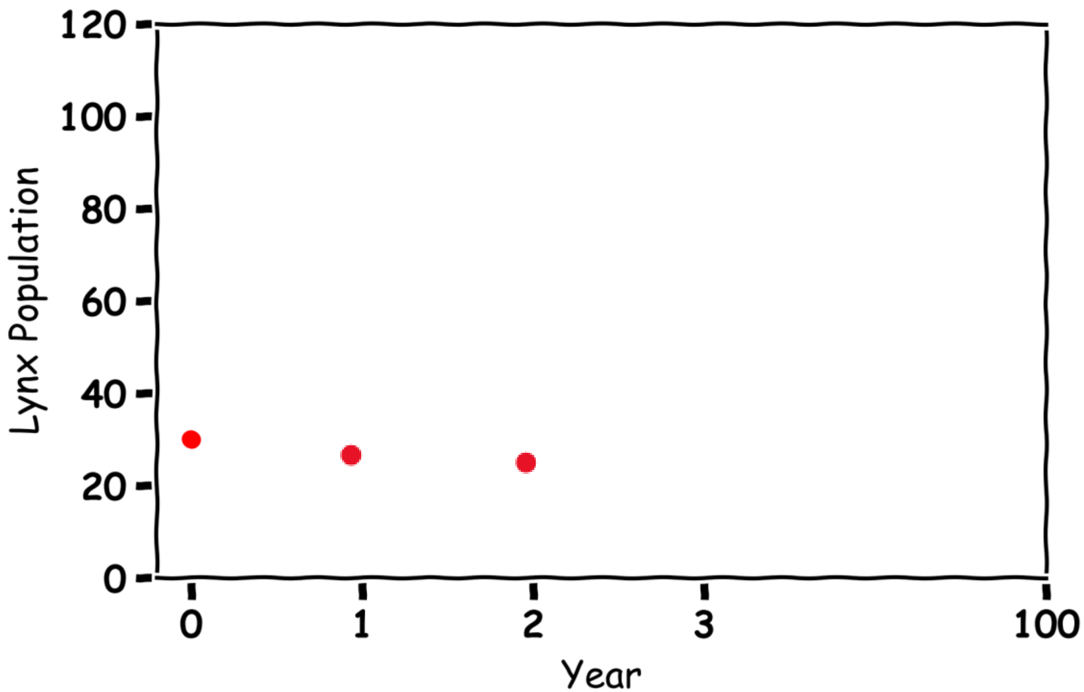
Exercise 4.: Predicting Three Years



$$L_{Future} = L - 0.1(L)$$



Year	Lynx Population
0	30
1	30-0.1(30)=27
2	27-0.1(27)=24
3	



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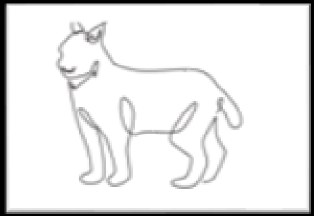
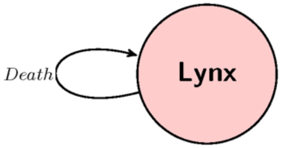
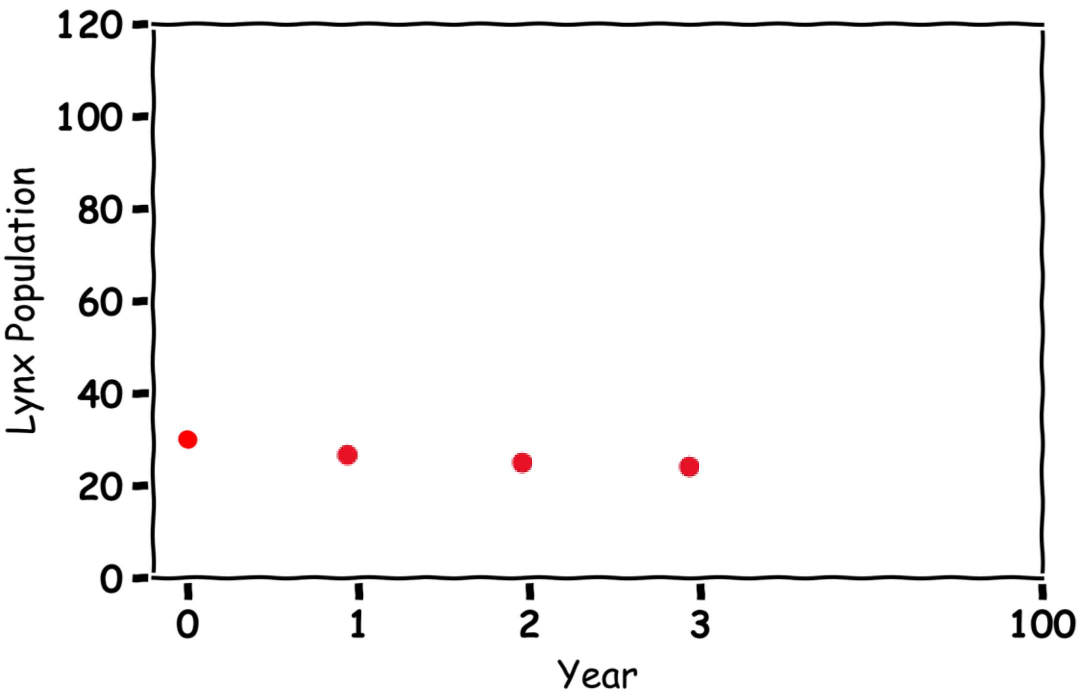
Exercise 5: Plotting Three Years



$$L_{Future} = L - 0.1(L)$$



Year	Lynx Population
0	30
1	30-0.1(30)=27
2	27-0.1(27)=24
3	24-0.1(24)=22



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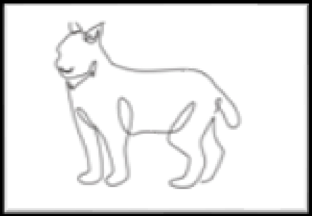
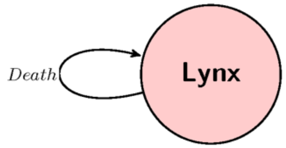
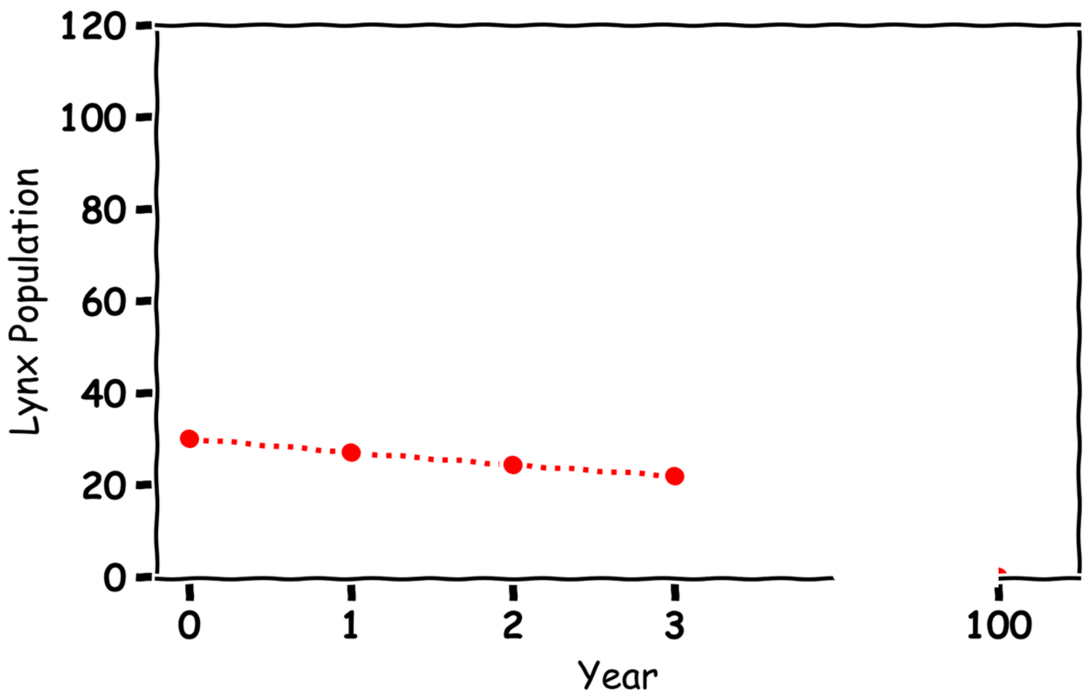
Exercise 5: Plotting Three Years

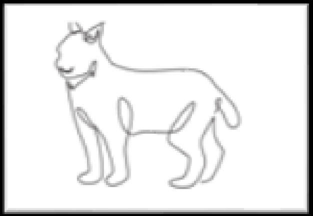


Year	Lynx Population
0	30
1	27
2	24
3	22

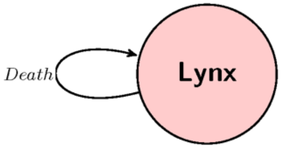


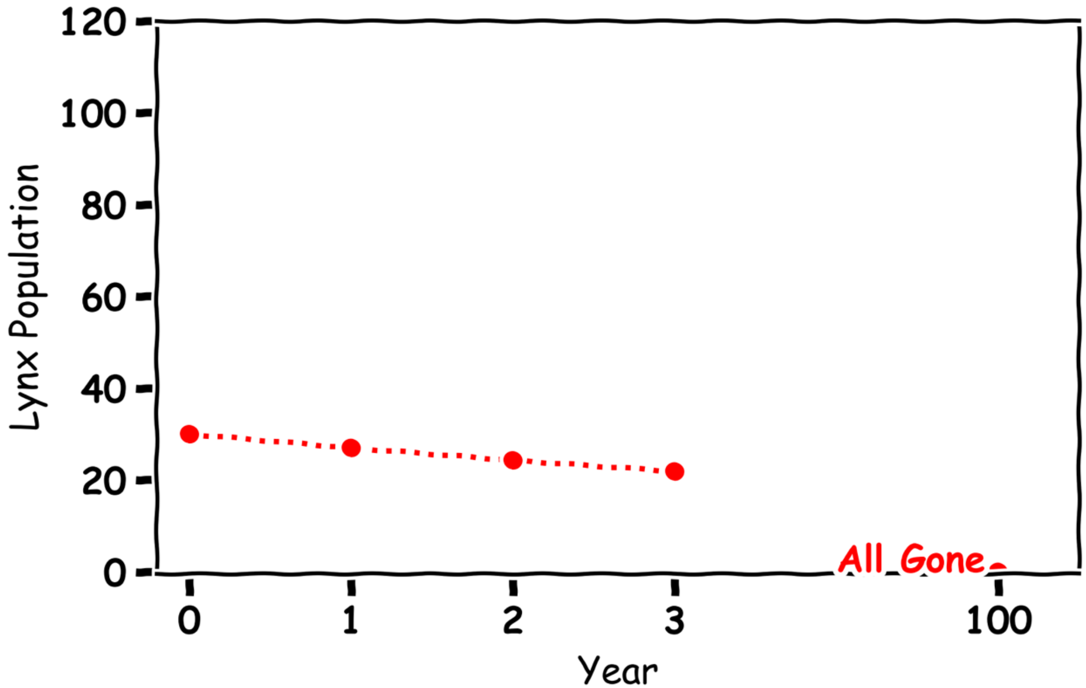


Exercise 6: Predicting 100 Years

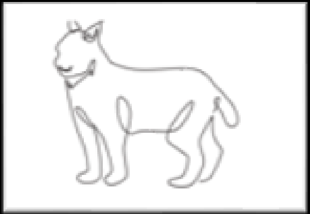


Year	Lynx Population
0	30
1	27
2	24
3	22





Lynx



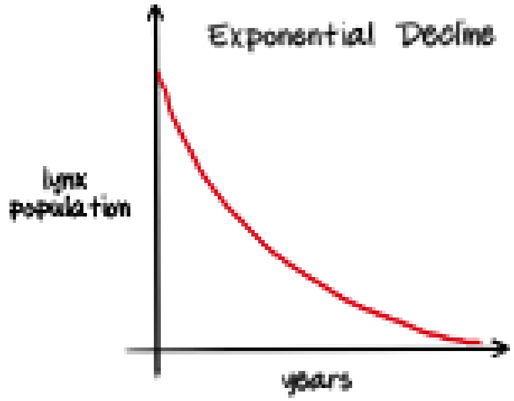
Ⓑ Lynx Population

L_c : current lynx population L_F : Future lynx population

$$r_{death} = \text{Deaths} - \text{Births}$$

$$\text{change in lynx population} = r_{death} \times L_c$$

$$L_F = L_c - \text{change in lynx population}$$

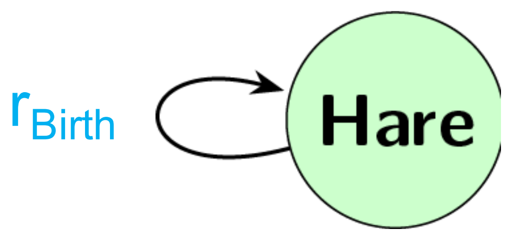


Predator and Prey

Predicting Future Hare Populations with Lynxes

$$\text{Future Hare Population} = \text{Current Hare Population} + \text{Birth Rate} * \text{Current Hare Population}$$

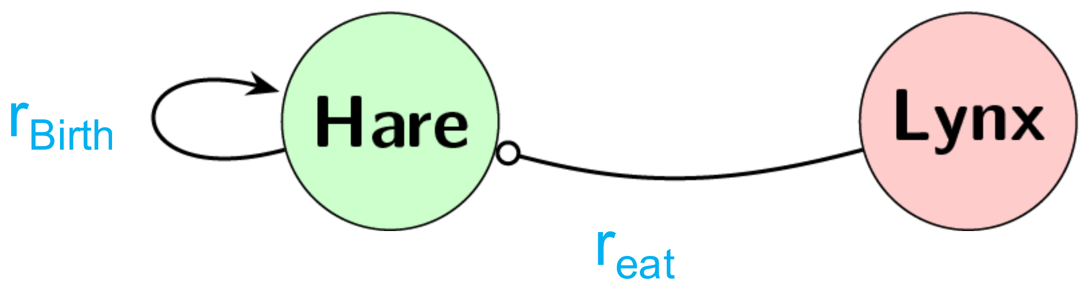
$$H = H + r_{Birth} H$$



Predicting Future Hare Populations with Lynxes

Future Hare Population = Hare Population + Birth Rate * Hare Population + Eat Rate * Hare Population * Lynx Population

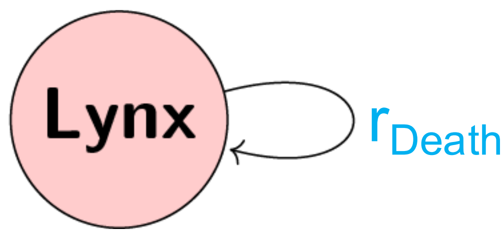
$$H=H+r_{Birth}H-r_{eat}LH$$



Predicting Future Hare Populations with Lynxes

$$\text{Future Lynx Population} = \text{Current Lynx Population} - \text{Death Rate} * \text{Current Lynx Population}$$

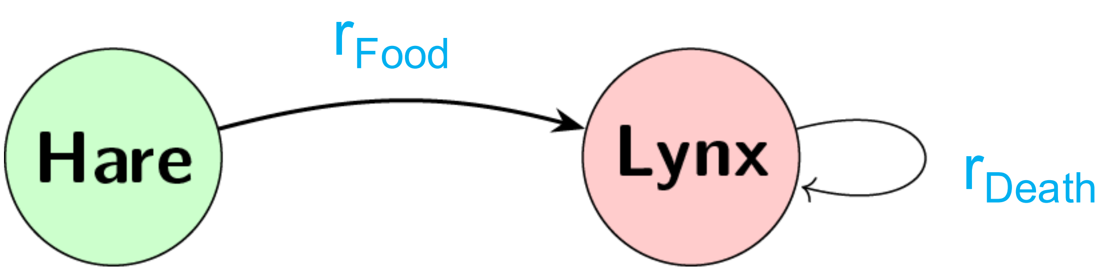
$$L = L - r_{Death} L$$



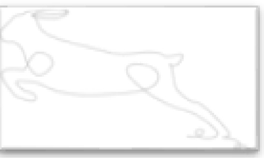
Predicting Future Lynx Populations with Hares

$$\text{Future Lynx Population} = \text{Lynx Population} - \text{Death Rate} * \text{Lynx Population} + \text{Food Rate} * \text{Hare Population} * \text{Lynx Population}$$

$$L = L - r_{Death} L + r_{Food} L H$$

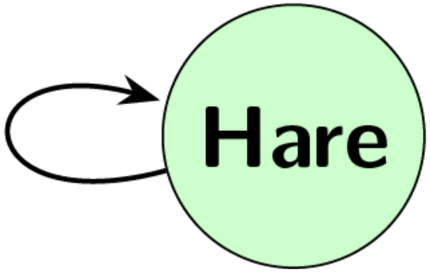


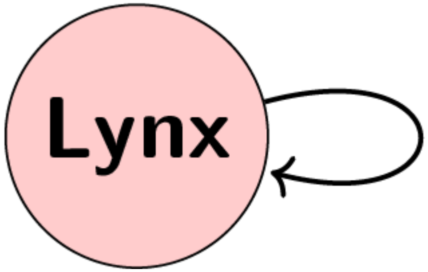
Predicting Future Hare and Lynx Populations



Hare

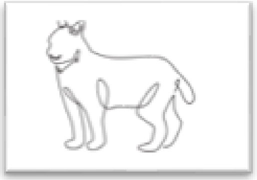
$H = H + r_{Birth} * H$



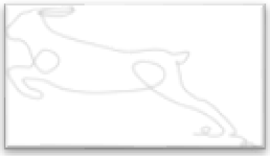


Lynx

$L = L - r_{Death} * L$

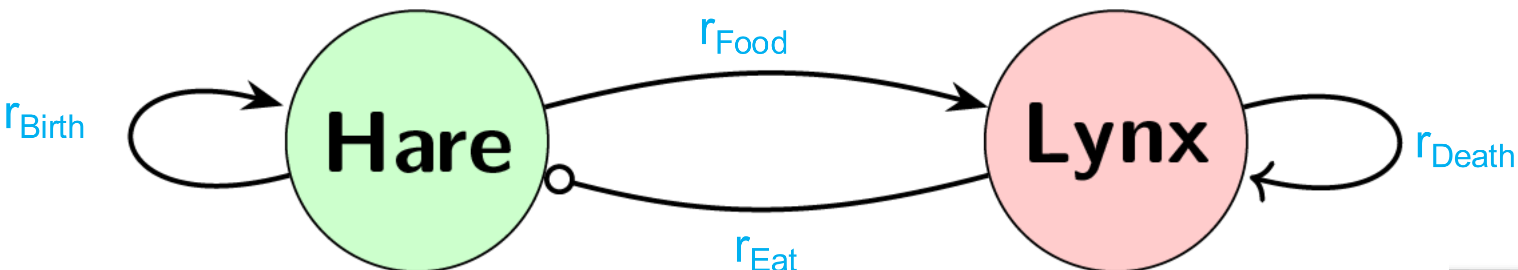


Predicting Many Years in the Future a year at a time



Hare

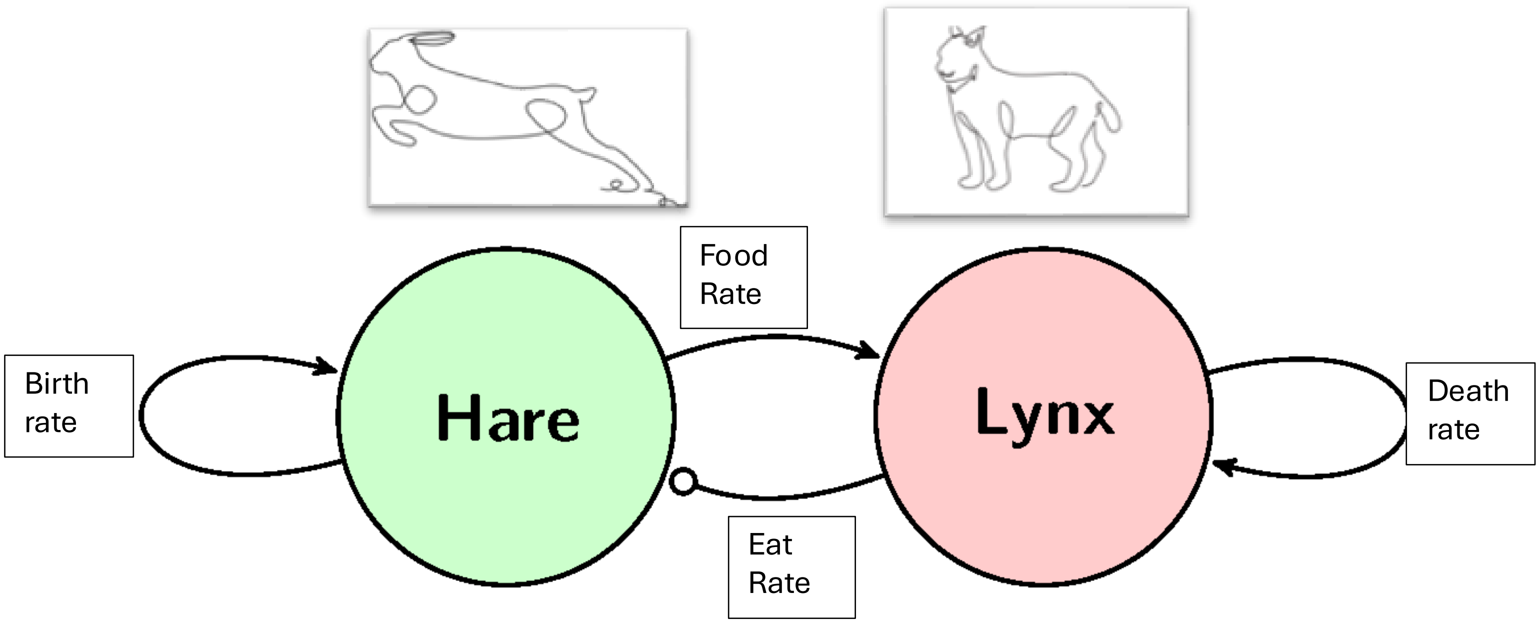
$$H = H + (r_{Birth} * H - r_{Eat} * H * L)$$



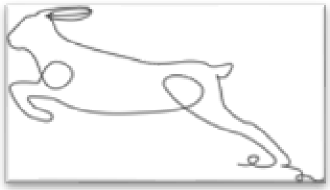
Lynx

$$L = L + (-r_{Death} * L + r_{Food} * H * L)$$

Exercise 7: Label the Graph



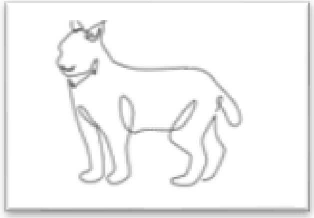
Predator and Prey



HARE POPULATION EQUATION

$$H_{Future} = H_{Current} + r_{Growth} \times H_{Current} - r_{Eat} \times H_{Current} \times L_{Current}$$

FUTURE HARE POPULATION = CURRENT HARE POPULATION + GROWTH RATE TIMES HARE POP-ULATION - EAT RATE OF HARE POP-ULATION BY LYNX POPULATION

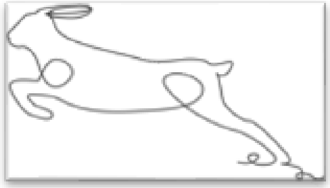


LYNX POPULATION EQUATION

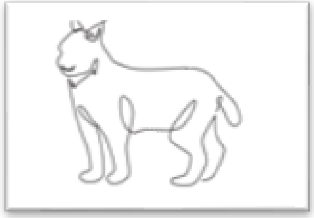
$$L_{Future} = L_{Current} - r_{Death} \times L_{Current} + r_{Food} \times L_{Current} \times H_{Current}$$

FUTURE LYNX POPULATION = CURRENT LYNX POPULATION - DEATH RATE TIMES LYNX POPULATION + FOOD RATE OF LYNX POPULATION BY HARE POP-ULATION

Predator and Prey



$$H_{Future} = H + 0.2(H) - 0.005(H)(L)$$



$$L_{Future} = L - 0.1(L) + 0.002(L)(H)$$

Exercise 8.1: Predicting One Year

$H_{Future}=H+0.2(H)-0.005(H)(L)$

$L_{Future}= L-0.1(L)+0.002(L)(H)$

Year	Hare Population	Lynx Population
0	60	30
1		
2		
3		

Exercise 8.1: Predicting One Year

$H_{Future} = H + 0.2(H) - 0.005(H)(L)$

$L_{Future} = L - 0.1(L) + 0.002(L)(H)$

Year	Hare Population	Lynx Population
0	60	30
1	60+0.2(60)-0.005(60)(30)=**63**	30-0.1(30)+0.002(30)(60)=**30.6**
2		
3		

Exercise 8.2: Predicting Three Years

$H_{Future} = H + 0.2(H) - 0.005(H)(L)$

$L_{Future} = L - 0.1(L) + 0.002(L)(H)$

Year	Hare Population	Lynx Population
0	60	30
1	60+0.2(60)-0.005(60)(30)=**63**	30-0.1(30)+0.002(30)(60)=**30.6**
2	63+0.2(63)-0.005(63)(30.6)=**65.9**	30.6-0.1(30.6)+0.002(30.6)(63)=**31.3**
3		

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Exercise 8.2: Predicting Three Years

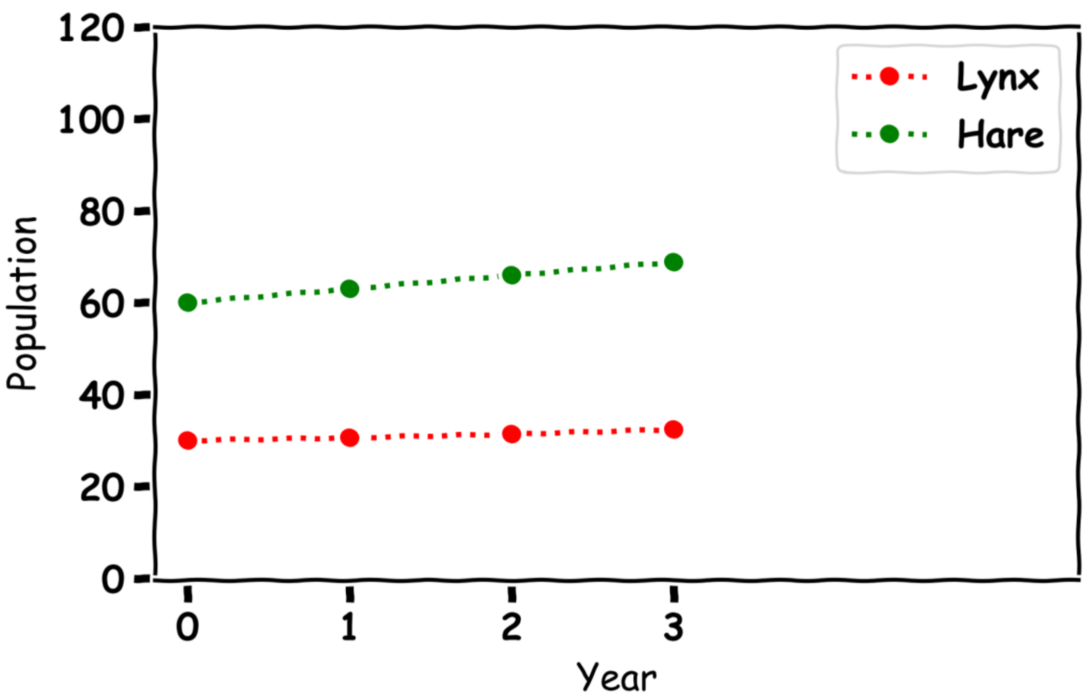
$H_{Future}=H+0.2(H)-0.005(H)(L)$

$L_{Future}= L-0.1(L)+0.002(L)(H)$

Year	Hare Population	Lynx Population
0	60	30
1	60+0.2(60)-0.005(60)(30)=**63**	30-0.1(30)+0.002(30)(60)=**30.6**
2	63+0.2(63)-0.005(63)(30.6)=**65.9**	30.6-0.1(30.6)+0.002(30.6)(63)=**31.3**
3	65.9+0.2(65.9)-0.005(65.9)(31.3)=**68.8**	31.3-0.1(31.3)+0.002(31.3)(65.9)=**32.4**

Exercise 9: Plotting Three Years

Year	Hare Population	Lynx Population
0	60	30
1	63	30.6
2	65.9	31.3
3	68.8	32.4



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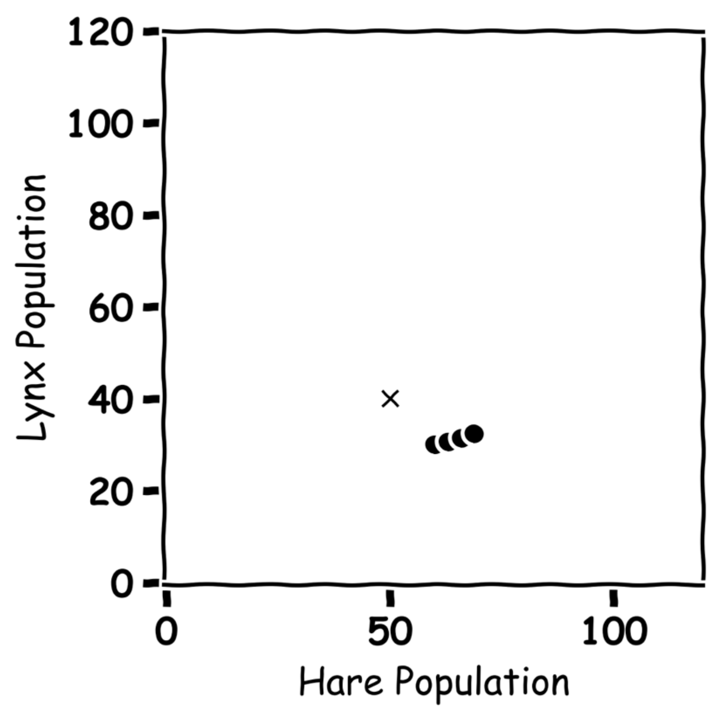
Exercise 10: Plotting Hares and Lynxes

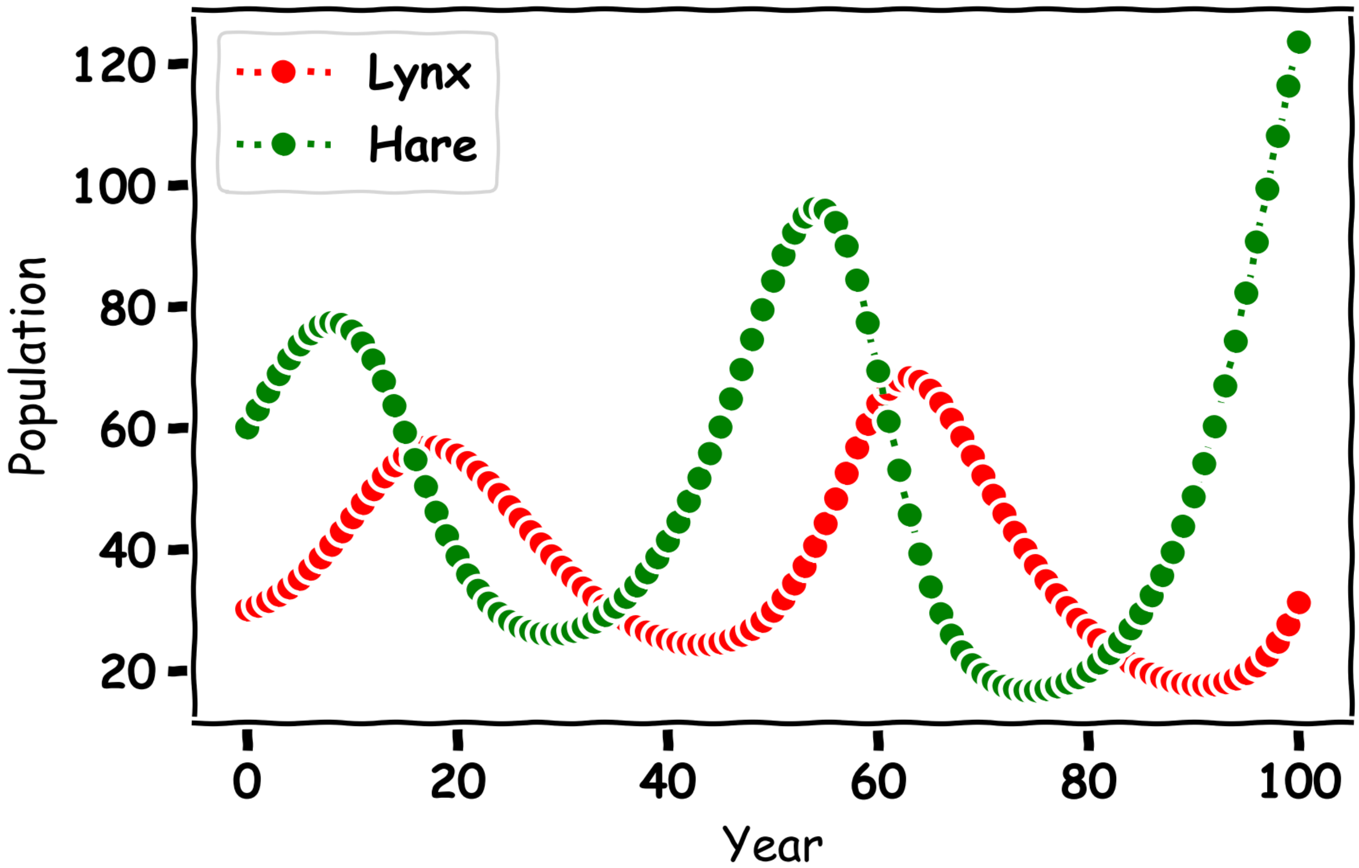
Year	Hare Population	Lynx Population
0	60	30
1		
2		
3		

Exercise 10: Plotting Hares and Lynxes

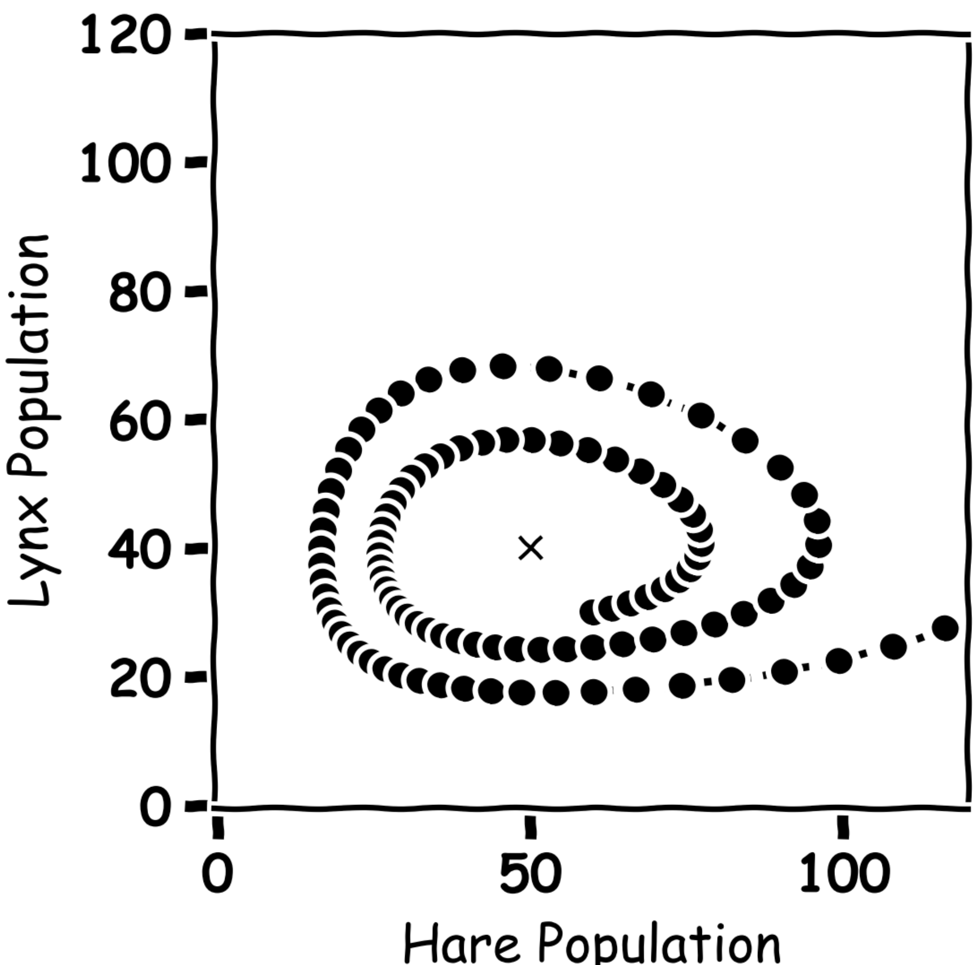
Year	Hare Population	Lynx Population
0	60	30
1	63	30.6
2	65.9	31.3
3	68.8	32.4



Predator and Prey

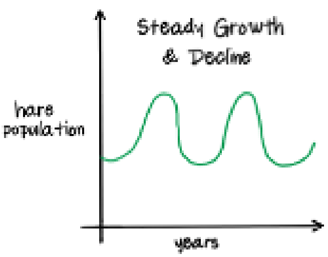


Predator and Prey



Predator and Prey

© Hare - Lynx Interaction



Hare Equation

r_{eaten} = hares lost to hunting

hares eaten = $r_{eaten} \times H_0 \times L_0$

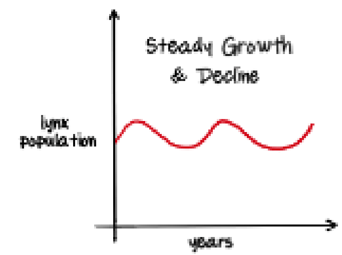
$H_F = H_0$ + change in hare population - hares eaten

Lynx Equation

r_{food} = successful lynx hunts

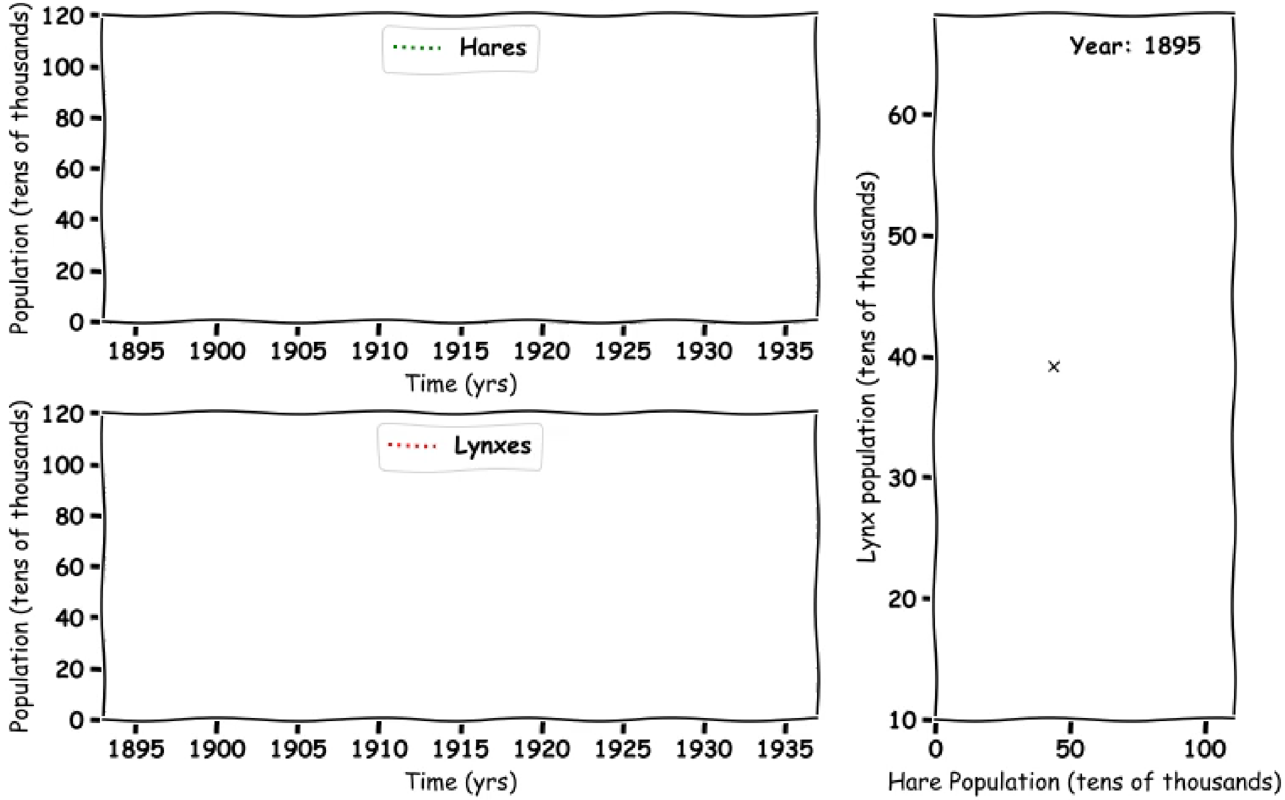
lynxes fed = $r_{food} \times H_0 \times L_0$

$L_F = L_0$ - change in lynx population + lynxes fed



Predator and Prey

The figure below plots the numerical approximation
of the Hare population (green) and the Lynx poulation (red)
as a function of time in years between 1895 and 1935.



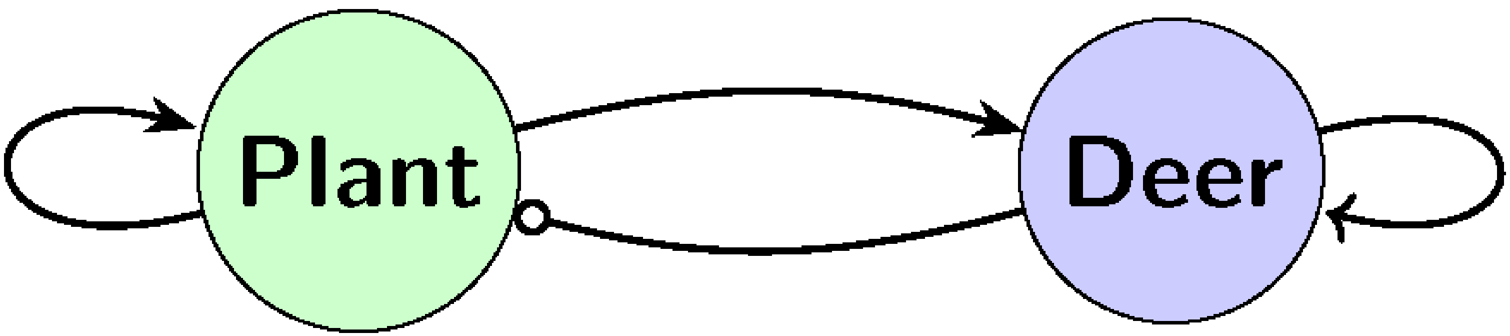
Predator Prey

Deer, Plants and Wolves

Plants and Deer

Plants

$$P[i+1]=P[i]+0.9*P[i]-0.02*P[i]*D[i])$$

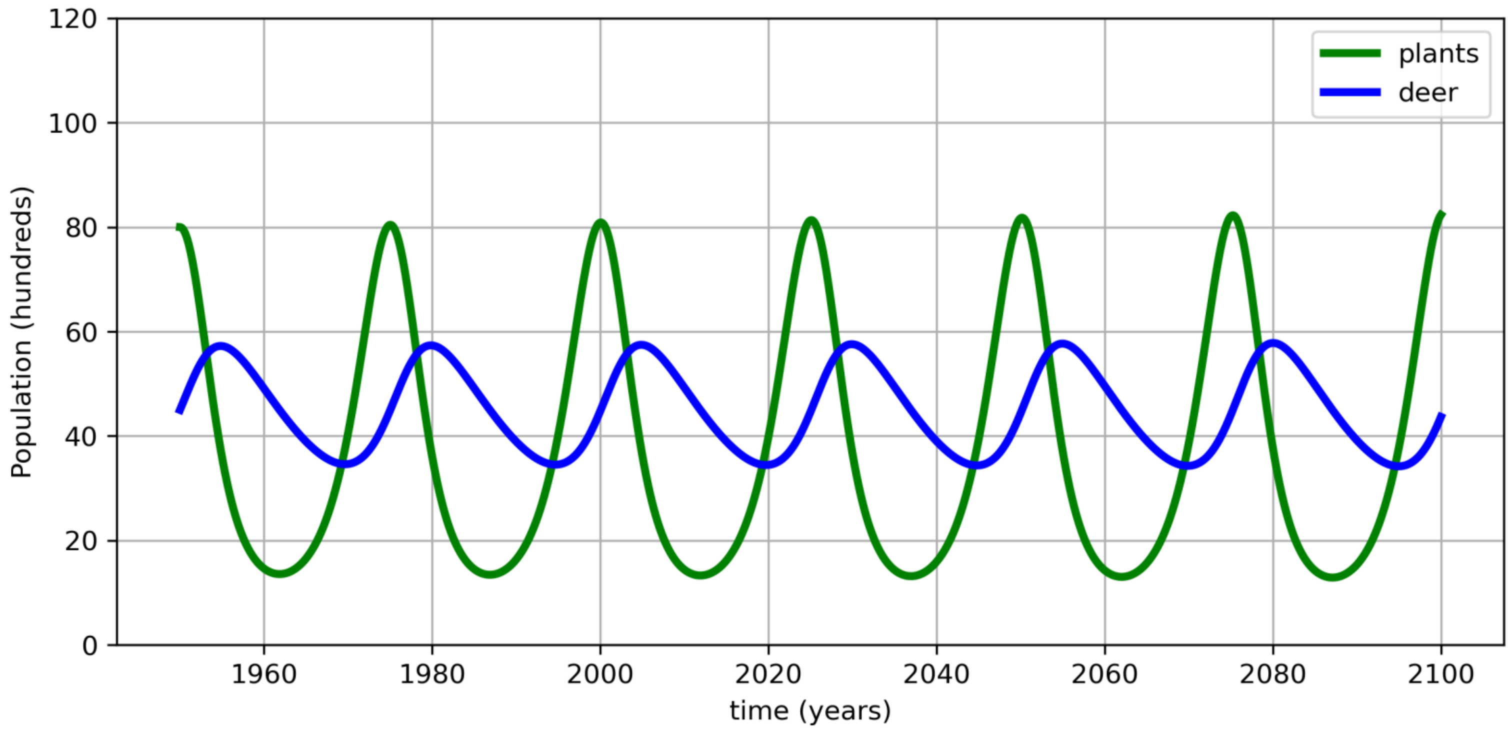


Deer

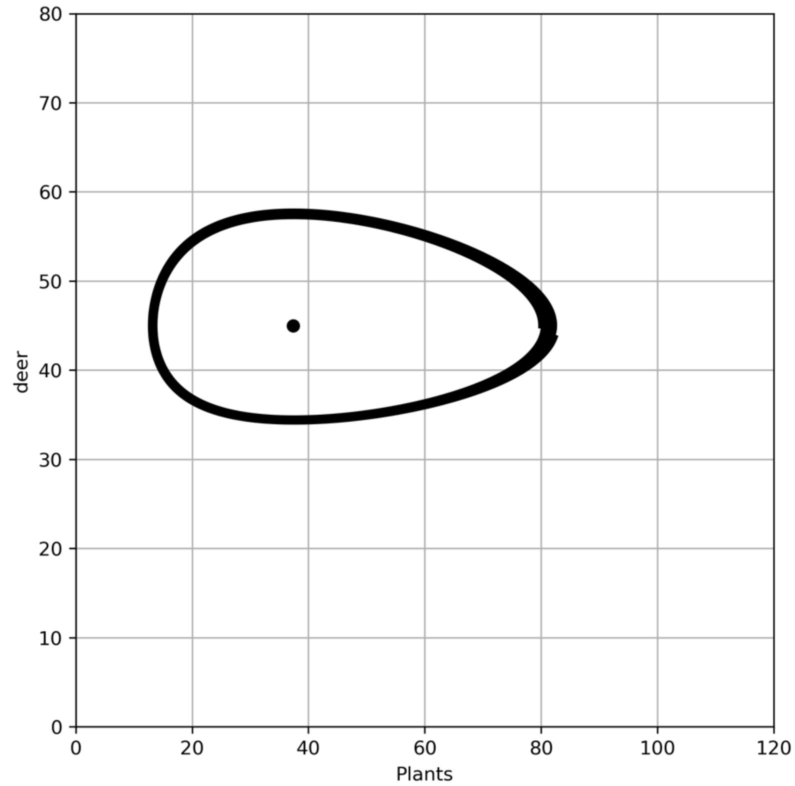
$$D[i+1]=D[i]-0.075*D[i]+0.002*P[i]*D[i]$$

$i=0,1,2,\ldots,N-1$

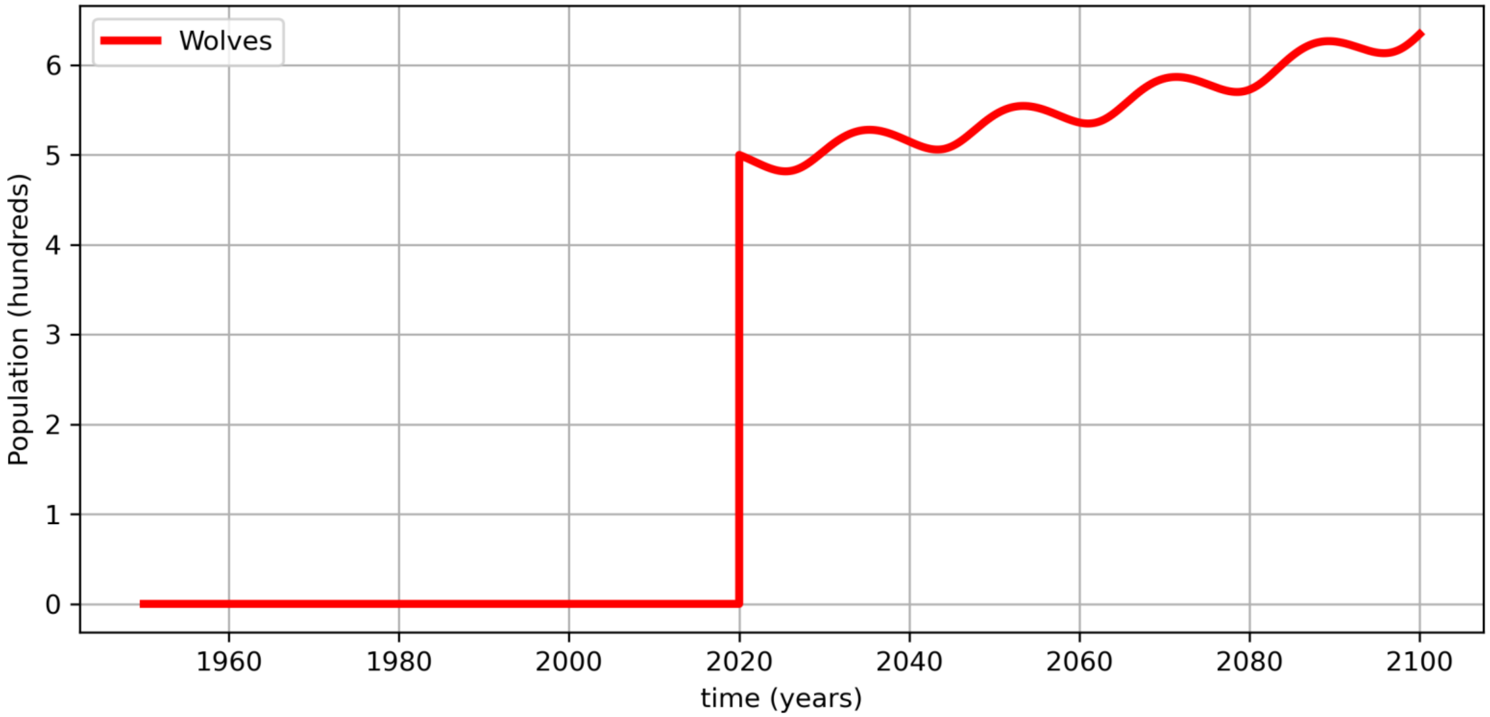
Plants and Deer



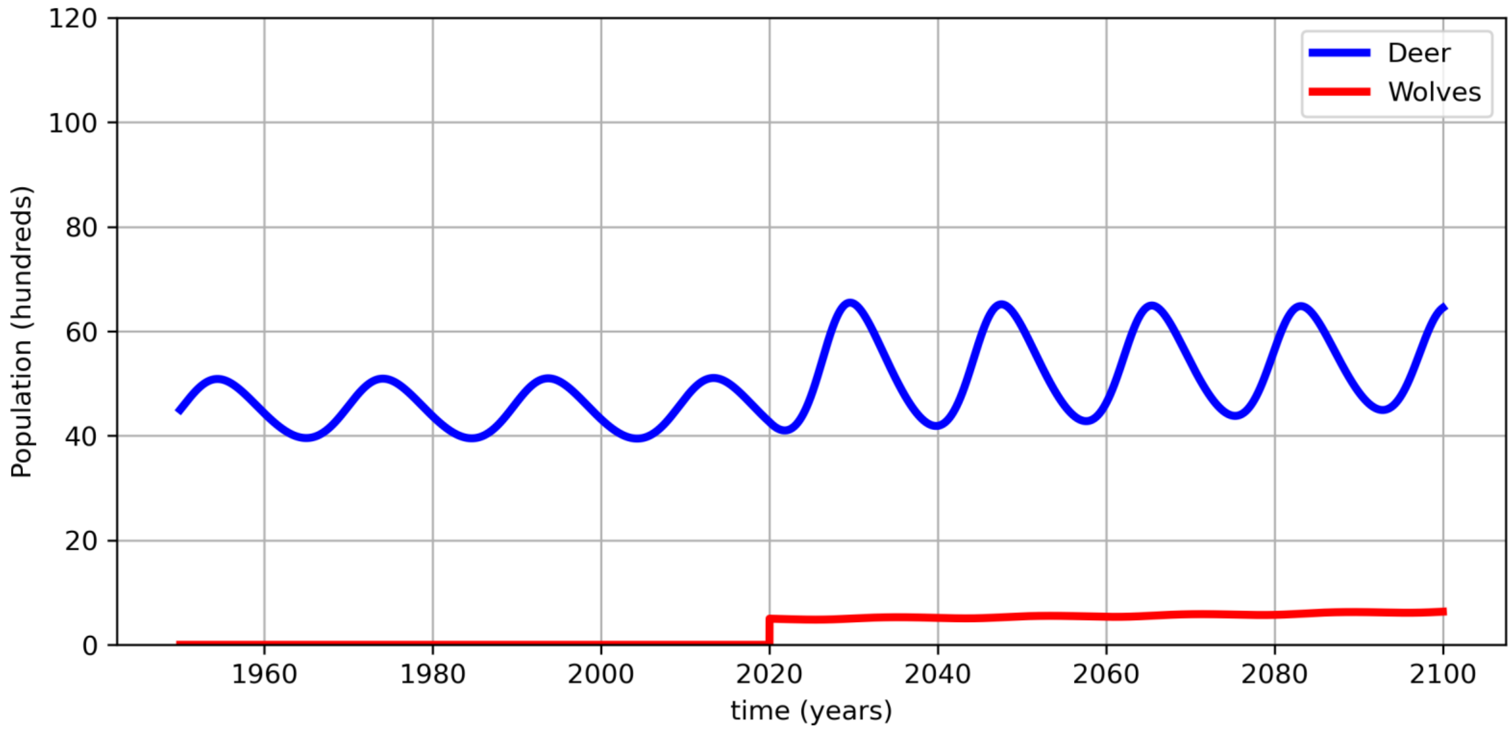
Deer vs Plant



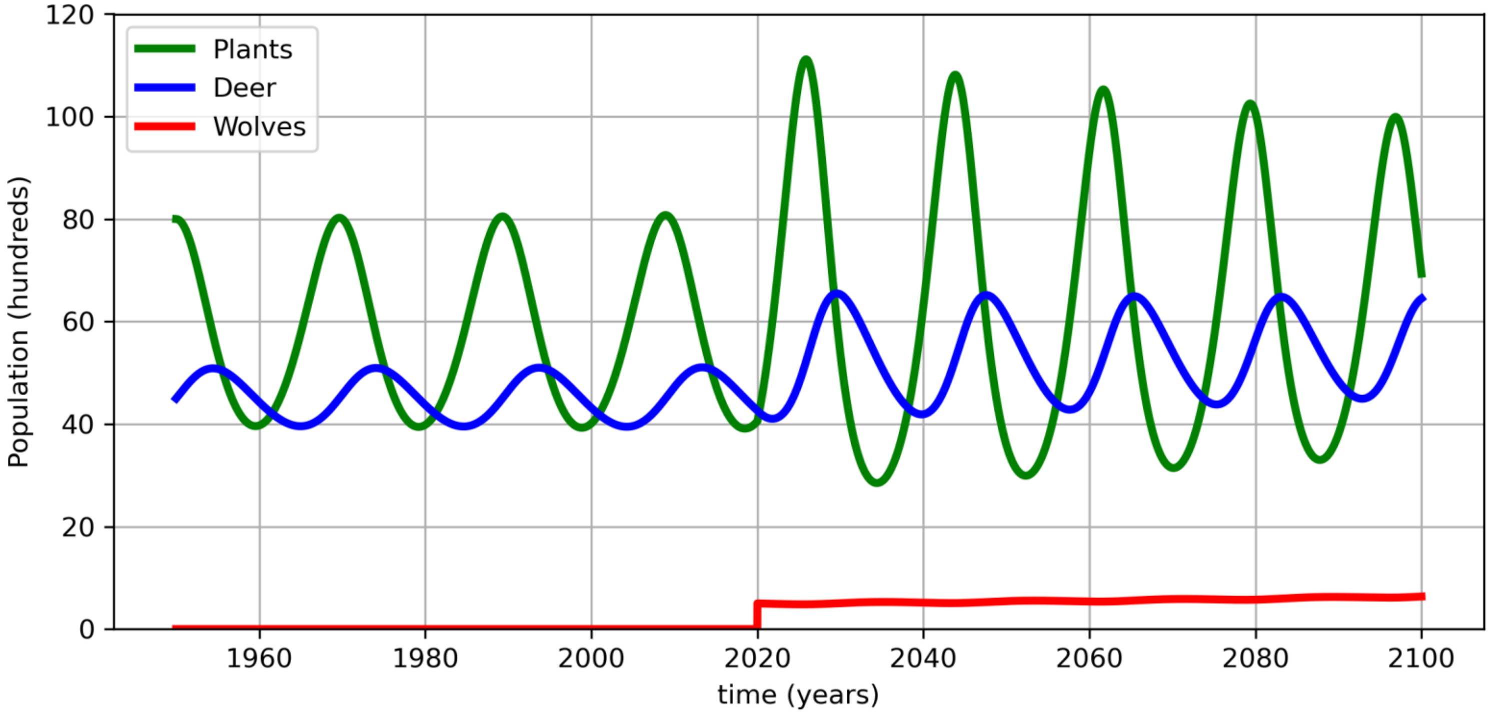
Wolves


Wolves
6
5
4
3
2
1
0
Population (hundreds)
1960
1980
2000
2020
2040
2060
2080
2100
time (years)

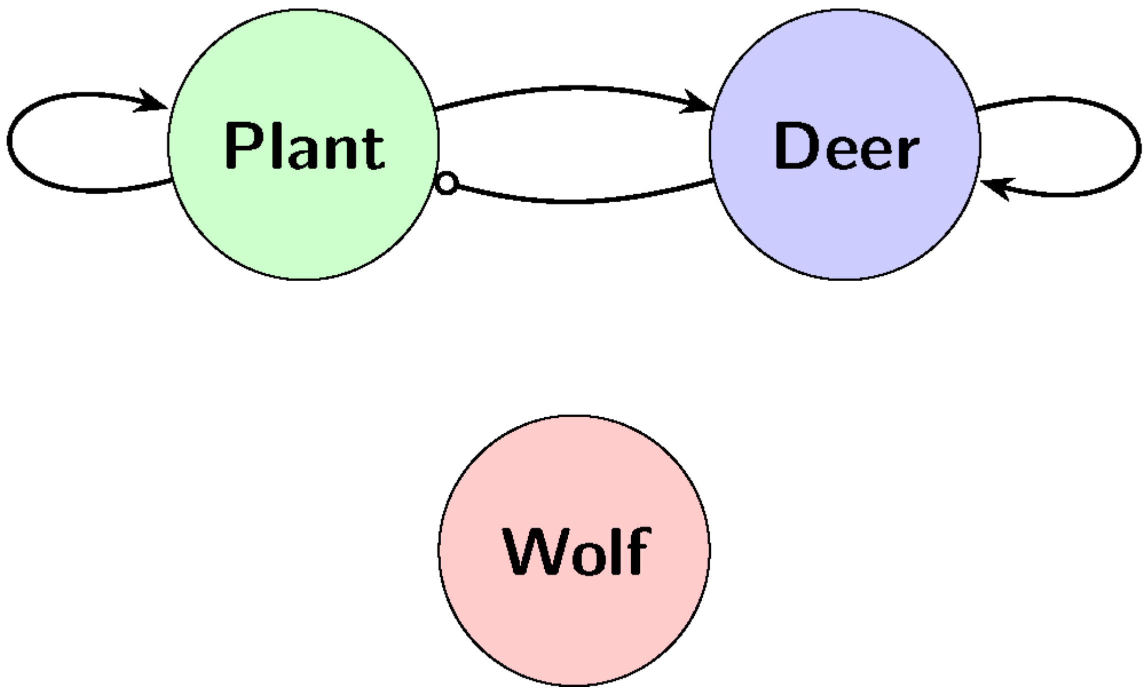
Wolves and Deer



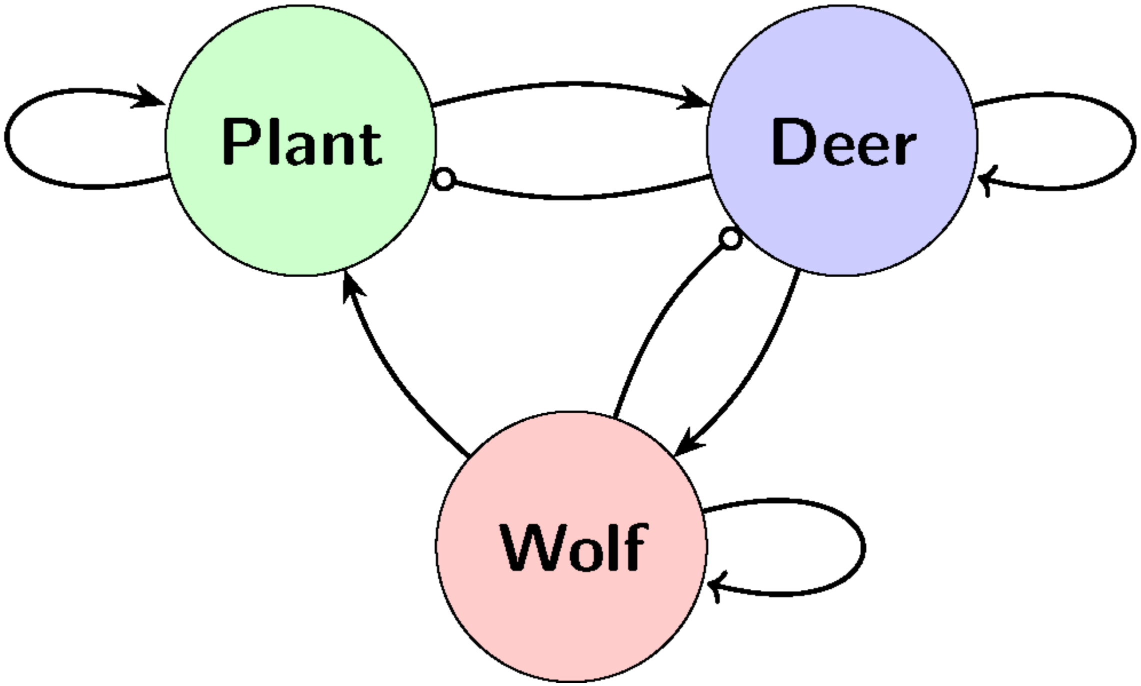
Wolves and Deer and Plants



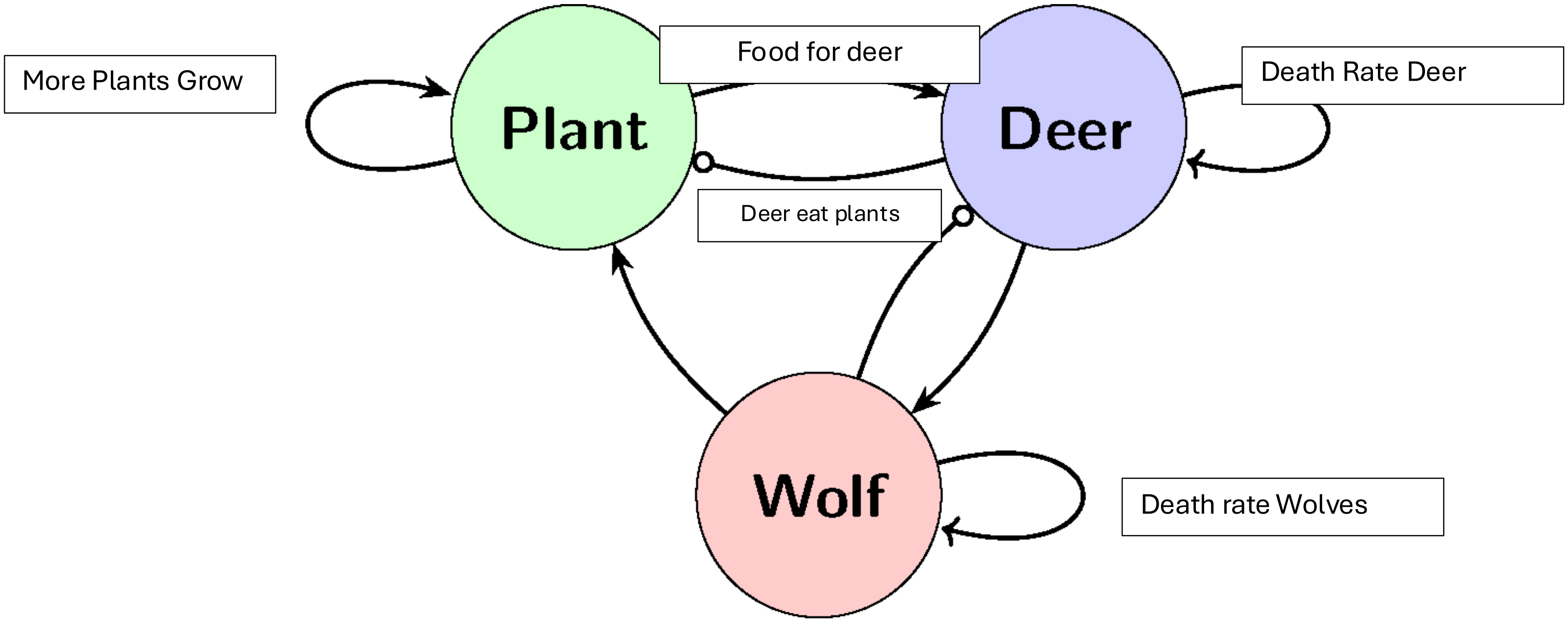
Directed Graph



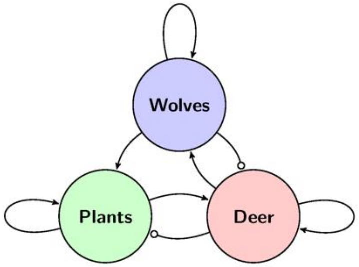
Exercise 11: Label the Graph



Directed Graph



Three Animals



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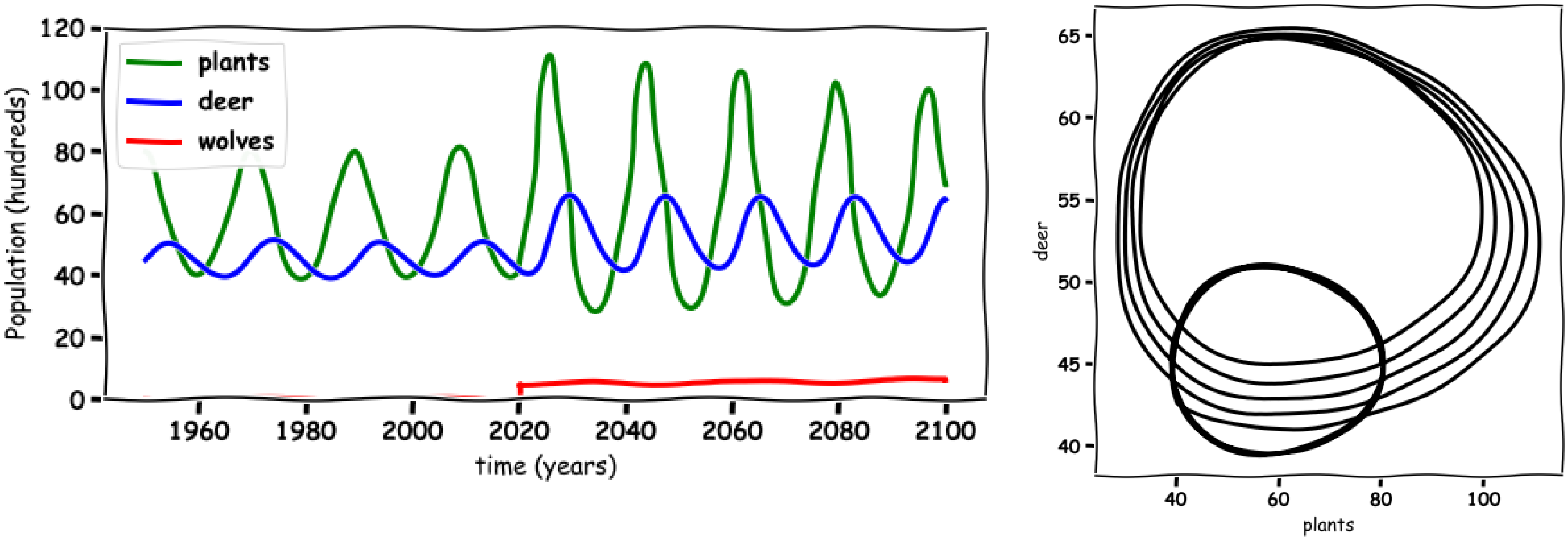
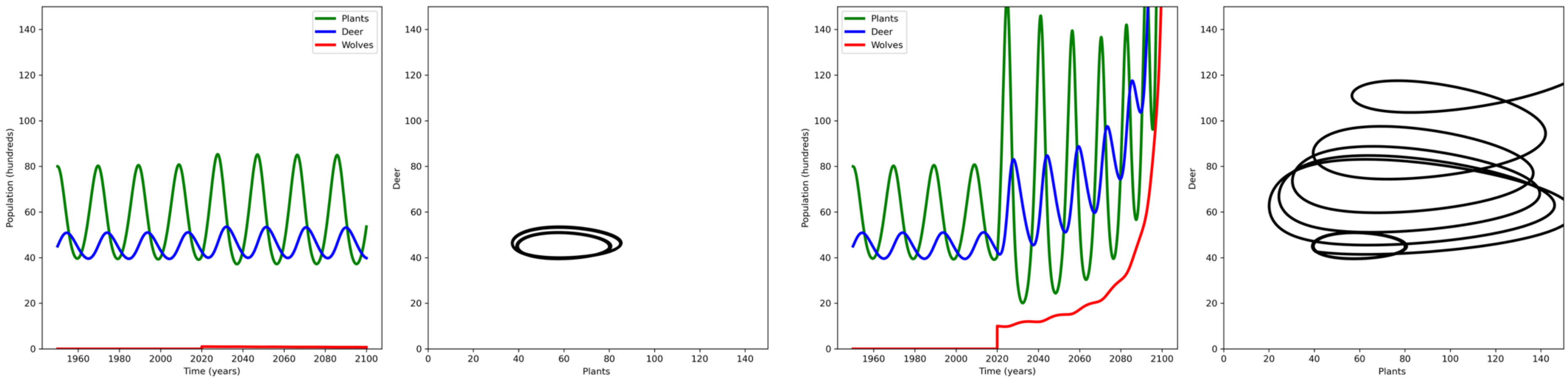
Exercise 12: Describe what is happening in the below graph



Do we have Enough Wolves



Discussion Questions

1. What happened to the environment when the wolves are re-introduced?

2. What happened to the wolf population when they are re-introduced?

3. What happened to the deer population when the wolves are re-introduced?

4. Do you think the wolves should have been re-introduced in Ireland?

5. What else could you model with these Equations?

Thank you for Listening

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